Group Optimization Report

Objective (aggregated fitness): 0.037405
Penalty (total aggregated): 0
Algorithm: Jde
Population size: 32
Parameter vector: 32 params (shared[0-10], Bas_2[11-17], Bas_1[18-24], Bas_0[25-31])
Input file: propellants.01234.json
Generated: 2026-08-07 17:49

Run Configuration (as resolved)

Provenance

- **Mode:** forward-eval
- **Configuration source:** /root/runs/3h-Classic-Tm2300-kc400-constrON-fsEquilibrium-TporeTs200-20260807-1132/run-config.json
- **Working directory:** /root/runs/3h-Classic-Tm2300-kc400-constrON-fsEquilibrium-TporeTs200-20260807-1132
- **Input propellants file:** propellants.01234.json
- **Resolved sidecar:** run_configuration.resolved.json
- **Recorded at (UTC):** 2026-08-07 17:49:29

Model constants

- **Metal melting temperature, K:** 2300
- **Surface-temperature search bracket, K:** 600 .. 900
- **Skeleton conduction contact factor:** 400 (contact-area fraction 0.0025; fixed calibration, not fitted)
- **Skeleton surface fraction:** equilibrium carbon: f_s = □ _Al+ □ _C(s)(T_pore,p) by Delesse, T_pore = (T_s + T_m)/2 — no fitted coefficients, no agglomeration data

coverage table: skeleton_carbon_equilibrium.json

Search

- **Strategy:** Classic
- **Parameter vector dimension:** 32
- **Population size (effective):** 1
- **Worker processors (effective):** 1 of 36 machine cores
- **RNG seed:** unseeded (run not exactly reproducible)
- **Termination:** not applicable (single point evaluation, no search)
- **Mutation force F:** 0.5
- **Crossover probability CR:** 0.9

Nelder-Mead refinement

- **Enabled:** yes
- **Stages:** final polish (100,000 evals)
- **Adaptive coefficients:** yes
- **Domain tolerance:** 1E-08
- **Function tolerance:** 1E-08
- **Restarts:** 2

Penalty thresholds

- **Penalty rate:** 0.01
- **Pocket heat-flux ratio threshold:** 100
- **Pore diameter threshold:** 3
- **Large oxidiser particle size threshold:** 1
- **Max inter-pocket kinetic-flame heat flux, W/m2:** 1E+09
- **Max skeleton kinetic-flame heat flux, W/m2:** 1E+08
- **Max out-skeleton kinetic-flame heat flux, W/m2:** 1E+08

Parameter bounds (18 base parameters)

- **ADecompose:** [1, 1E+15]

EDecompose: [50000, 300000]
AKineticFlameInterPocket: [100000, 1E+13]
EKineticFlameInterPocket: [50000, 250000]
AKineticFlamePocketOutSkeleton: [100000, 1E+13]
EKineticFlamePocketOutSkeleton: [50000, 250000]
AKineticFlamePocketSkeleton: [100000, 1E+13]
EKineticFlamePocketSkeleton: [50000, 250000]
NuInterPocket: [0, 2.5]
NuPocketOutSkeleton: [0, 2.5]
NuPocketSkeleton: [0, 2.5]
AMetalBurningConstant: [1E-10, 0.001]
BMetalBurningConstant: [1E-12, 1E-05]
DeltaH: [-1E+07, 1E+07]
KDiffusionHeight: [0.001, 10]
APowOrder: [0, 3]
BPowOrder: [0, 3]
KCoefficientRadiationTemperature: [0, 1]

Functional Constraints
Flame competition constraint within pocket regions.
Penalty coefficient 0.0100
Maximum allowable heat flux density ratio (max/min) 100.00
The linear burning rate of the first conditional fuel is greater than that of the second.
Penalty coefficient 0.0100
Constraint on maximum kinetic flame heat flux density.
Penalty coefficient 0.0100
Maximum kinetic flame heat flux density in inter-pocket medium 1.00e+09
Maximum kinetic flame heat flux density in skeleton structure 1.00e+08
Maximum kinetic flame heat flux density in pocket region outside skeleton 1.00e+08
Constraint on skeleton thickness to pore diameter ratio.
Penalty coefficient 0.0100
Threshold for skeleton thickness to pore diameter ratio 3.00
Constraint on skeleton thickness to large oxidizer particle diameter ratio.
Penalty coefficient 0.0100
Threshold for skeleton thickness to large oxidizer particle diameter ratio 1.00

Parametric Constraints
Constraint on allowable surface temperature range
Minimum surface temperature 600 K
Maximum surface temperature 900 K

Differential Evolution Algorithm
Population size: 32
Mutation force: 0.5
Crossover probability: 0.9
Number of threads: 1
Termination strategy: TimeoutTerminationStrategy

Group Combustion Solver Parameters
Values below are display-rounded — not for replay; use the Reproduction Vector block.
Each parameter shows its bounds and a rail flag: [RAILED@lo]/[RAILED@hi] means the gene sits within 0.1% of a bound (fit is bound-limited — consider widening); [interior] means it does not.
Shared Parameters (All Compositions)
Pre-exponential coefficient for gas phase reactions in inter-pocket medium 270425.16247224977
Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas phase reactions in inter-pocket medium 50000.01390619085
Parameter bounds [5.000E+004; 2.500E+005] [RAILED@lo]
Pre-exponential factor for gas-phase reactions in pocket region outside skeleton 835575.1158096384

Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions in pocket region outside skeleton 50000.000048000475
Parameter bounds [5.000E+004; 2.500E+005] [RAILED@lo]
Reaction order for gas-phase chemistry in inter-pocket medium 2.4999999994170774
Parameter bounds [0.000E+000; 2.500E+000] [RAILED@hi]
Reaction order for gas-phase chemistry outside skeleton structure 1.2600393129619831
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Enthalpy difference between vaporization and binder decomposition -419917.58660087275
Parameter bounds [-1.000E+007; 1.000E+007] [interior]
Diffusion flame height correlation coefficient 1.3136
Parameter bounds [1.000E-003; 1.000E+001] [interior]
Power law exponent for burning rate dependency on A (skeleton thickness) 1.4146
Parameter bounds [0.000E+000; 3.000E+000] [interior]
Power law exponent for burning rate dependency on B (pore diameter) 0.5896
Parameter bounds [0.000E+000; 3.000E+000] [interior]
Radiation temperature correlation coefficient (average layer temperature) 0.0000
Parameter bounds [0.000E+000; 1.000E+000] [RAILED@lo]

Bas_2 (includes Bas_3, Bas_4) - Condensed Phase Parameters
Pre-exponential coefficient for condensed phase reactions 149453556477435.6
Parameter bounds [1.000E+000; 1.000E+015] [interior]
Activation energy for condensed phase reactions 153149.85546090113
Parameter bounds [5.000E+004; 3.000E+005] [interior]
Pre-exponential factor for gas-phase reactions within skeleton structure 199782.74876397804
Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions within skeleton structure 50000.0000505755
Parameter bounds [5.000E+004; 2.500E+005] [RAILED@lo]
Reaction order for gas-phase chemistry in skeleton structure 2.3952110084855476
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Coefficient A for metal combustion heat flux density in skeleton structure, 4.21285981107954E-08
Parameter bounds [1.000E-010; 1.000E-003] [RAILED@lo]
Coefficient B for metal combustion heat flux density in skeleton structure, 2.6409363928164703E-07
Parameter bounds [1.000E-012; 1.000E-005] [RAILED@lo]

Bas_1 - Condensed Phase Parameters
Pre-exponential coefficient for condensed phase reactions 999999746557452.5
Parameter bounds [1.000E+000; 1.000E+015] [RAILED@hi]
Activation energy for condensed phase reactions 162164.04940057814
Parameter bounds [5.000E+004; 3.000E+005] [interior]
Pre-exponential factor for gas-phase reactions within skeleton structure 6082663369219.321
Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions within skeleton structure 209617.60464566582
Parameter bounds [5.000E+004; 2.500E+005] [interior]
Reaction order for gas-phase chemistry in skeleton structure 2.480302163720106
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Coefficient A for metal combustion heat flux density in skeleton structure, 1.1880525970877363E-08
Parameter bounds [1.000E-010; 1.000E-003] [RAILED@lo]
Coefficient B for metal combustion heat flux density in skeleton structure, 2.2969185280399496E-12
Parameter bounds [1.000E-012; 1.000E-005] [RAILED@lo]

Bas_0 - Condensed Phase Parameters
Pre-exponential coefficient for condensed phase reactions 999999727101067.5
Parameter bounds [1.000E+000; 1.000E+015] [RAILED@hi]
Activation energy for condensed phase reactions 226703.19299759885
Parameter bounds [5.000E+004; 3.000E+005] [interior]
Pre-exponential factor for gas-phase reactions within skeleton structure 8773525.267853536
Parameter bounds [1.000E+005; 1.000E+013] [interior]
Activation energy for gas-phase reactions within skeleton structure 50000.03248774392
Parameter bounds [5.000E+004; 2.500E+005] [RAILED@lo]

Reaction order for gas-phase chemistry in skeleton structure 0.5054928018529032
Parameter bounds [0.000E+000; 2.500E+000] [interior]
Coefficient A for metal combustion heat flux density in skeleton structure, 2.0796104933579726E-08
Parameter bounds [1.000E-010; 1.000E-003] [RAILED@lo]
Coefficient B for metal combustion heat flux density in skeleton structure, 3.678888478168093E-07
Parameter bounds [1.000E-012; 1.000E-005] [RAILED@lo]

<u>Reproduction Vector</u>

Full-precision result vector. Copy the block below (including the '#' lines) into a text file and replay this run exactly with: --forward-eval <file>


```
# reproduction vector (32 params)
# objective (aggregated fitness) = 0.03740504378415831
# penalty (total aggregated) = 0
# input file = propellants.01234.json
# population = 32
# generated = 2026-08-07 17:49
# layout = shared[0-10], Bas_2[11-17], Bas_1[18-24], Bas_0[25-31]
# count = 32
# checksum = 2b55f615e30dc44f008d49f159fe2d4a4714fca2da2ec24ba5c4ea6303e9c535
270425.16247224977
50000.01390619085
835575.1158096384
50000.000048000475
2.4999999994170774
1.2600393129619831
-419917.58660087275
1.3135664540421474
1.4145863782957049
0.589586784501796
1.6631647399145713E-05
149453556477435.6
153149.85546090113
199782.74876397804
50000.0000505755
2.3952110084855476
4.21285981107954E-08
2.6409363928164703E-07
999999746557452.5
162164.04940057814
6082663369219.321
209617.60464566582
2.480302163720106
1.1880525970877363E-08
2.2969185280399496E-12
999999727101067.5
226703.19299759885
8773525.267853536
50000.03248774392
0.5054928018529032
2.0796104933579726E-08
3.678888478168093E-07
```


Optimization Results

Objective function value 0.0374
Total penalty from constraint violations 0.0000

<u>Burn-Rate Error Breakdown</u>

*Per-point relative error e = (calc/exp - 1)*100%. A fuel's RMS is its contribution to the objective (sqrt of the mean squared fractional error over all pressures); a composition objective is the mean of its per-fuel RMS, and the overall objective is the mean of the three compositions. Numbers below reflect this report's input vector.*

<u>**Bas_2 (includes Bas_3, Bas_4) - Burn-Rate Errors**</u>

Bas_2

p = 1 MPa: exp 16.9441, calc 15.1542 mm/s -> err -10.56%
p = 1.611 MPa: exp 21.7132, calc 19.8929 mm/s -> err -8.38%
p = 2.222 MPa: exp 25.6654, calc 24.0886 mm/s -> err -6.14%
p = 2.833 MPa: exp 29.1215, calc 27.9415 mm/s -> err -4.05%
p = 3.444 MPa: exp 32.2345, calc 31.5541 mm/s -> err -2.11%
p = 4.056 MPa: exp 35.0917, calc 34.9879 mm/s -> err -0.30%
p = 4.667 MPa: exp 37.7487, calc 38.2828 mm/s -> err +1.41%
p = 5.278 MPa: exp 40.2432, calc 41.4668 mm/s -> err +3.04%
p = 5.889 MPa: exp 42.6026, calc 44.5604 mm/s -> err +4.60%
p = 6.5 MPa: exp 44.8470, calc 47.5785 mm/s -> err +6.09%
Per-fuel RMS = 0.0557 (5.57%)

Bas_3

p = 1 MPa: exp 15.2411, calc 15.4654 mm/s -> err +1.47%
p = 1.611 MPa: exp 20.6812, calc 20.3180 mm/s -> err -1.76%
p = 2.222 MPa: exp 25.4073, calc 24.7172 mm/s -> err -2.72%
p = 2.833 MPa: exp 29.6815, calc 28.9348 mm/s -> err -2.52%
p = 3.444 MPa: exp 33.6334, calc 33.0607 mm/s -> err -1.70%
p = 4.056 MPa: exp 37.3394, calc 37.1323 mm/s -> err -0.55%
p = 4.667 MPa: exp 40.8488, calc 41.1676 mm/s -> err +0.78%
p = 5.278 MPa: exp 44.1961, calc 45.1772 mm/s -> err +2.22%
p = 5.889 MPa: exp 47.4063, calc 49.1672 mm/s -> err +3.71%
p = 6.5 MPa: exp 50.4986, calc 53.1422 mm/s -> err +5.24%
Per-fuel RMS = 0.0262 (2.62%)

Bas_4

p = 1 MPa: exp 9.7279, calc 11.7883 mm/s -> err +21.18%
p = 1.611 MPa: exp 13.5834, calc 14.1190 mm/s -> err +3.94%
p = 2.222 MPa: exp 17.0126, calc 16.5100 mm/s -> err -2.95%
p = 2.833 MPa: exp 20.1664, calc 19.0542 mm/s -> err -5.51%
p = 3.444 MPa: exp 23.1208, calc 21.7263 mm/s -> err -6.03%
p = 4.056 MPa: exp 25.9212, calc 24.4929 mm/s -> err -5.51%
p = 4.667 MPa: exp 28.5972, calc 27.3290 mm/s -> err -4.43%
p = 5.278 MPa: exp 31.1699, calc 30.2171 mm/s -> err -3.06%
p = 5.889 MPa: exp 33.6545, calc 33.1456 mm/s -> err -1.51%
p = 6.5 MPa: exp 36.0627, calc 36.1062 mm/s -> err +0.12%
Per-fuel RMS = 0.0775 (7.75%)

Composition objective (mean per-fuel RMS) = 0.0532 (5.32%)

<u>**Bas_1 - Burn-Rate Errors**</u>

Bas_1

p = 1 MPa: exp 14.2647, calc 15.8080 mm/s -> err +10.82%
p = 1.611 MPa: exp 19.9183, calc 20.7398 mm/s -> err +4.12%
p = 2.222 MPa: exp 24.9468, calc 25.1329 mm/s -> err +0.75%
p = 2.833 MPa: exp 29.5714, calc 29.1969 mm/s -> err -1.27%
p = 3.444 MPa: exp 33.9037, calc 33.0386 mm/s -> err -2.55%
p = 4.056 MPa: exp 38.0101, calc 36.7216 mm/s -> err -3.39%
p = 4.667 MPa: exp 41.9342, calc 40.2871 mm/s -> err -3.93%
p = 5.278 MPa: exp 45.7067, calc 43.7631 mm/s -> err -4.25%
p = 5.889 MPa: exp 49.3500, calc 47.1713 mm/s -> err -4.41%
p = 6.5 MPa: exp 52.8815, calc 50.5257 mm/s -> err -4.45%
Per-fuel RMS = 0.0477 (4.77%)

Composition objective (mean per-fuel RMS) = 0.0477 (4.77%)

Bas_0 - Burn-Rate Errors

Bas_0

p = 1 MPa: exp 5.9659, calc 6.0463 mm/s -> err +1.35%
p = 1.611 MPa: exp 7.9046, calc 7.8409 mm/s -> err -0.81%
p = 2.222 MPa: exp 9.5561, calc 9.4280 mm/s -> err -1.34%
p = 2.833 MPa: exp 11.0289, calc 10.8960 mm/s -> err -1.21%
p = 3.444 MPa: exp 12.3759, calc 12.2778 mm/s -> err -0.79%
p = 4.056 MPa: exp 13.6278, calc 13.5914 mm/s -> err -0.27%
p = 4.667 MPa: exp 14.8044, calc 14.8481 mm/s -> err +0.30%
p = 5.278 MPa: exp 15.9192, calc 16.0562 mm/s -> err +0.86%
p = 5.889 MPa: exp 16.9822, calc 17.2220 mm/s -> err +1.41%
p = 6.5 MPa: exp 18.0009, calc 18.3504 mm/s -> err +1.94%
Per-fuel RMS = 0.0114 (1.14%)

Composition objective (mean per-fuel RMS) = 0.0114 (1.14%)

Overall objective (mean of the three compositions) = 0.0374 (3.74%)

Vieille Power-Law Fit (U = A*p^v, U in m/s, p in Pa)

*Experimental A/v are the supplied Vieille coefficients (the experimental burn rate is defined as exactly A*p^v, so they are exact, not re-fitted). Calculated A/v are an ordinary-least-squares fit of the model burn rates in log-log space; R2 shows how power-law-like the model curve is, and dv = v_calc - v_exp is the model's error in the pressure exponent. Numbers below reflect this report's input vector.*

Bas_2 (includes Bas_3, Bas_4) - Vieille Coefficients

Bas_2

exp: A = 1.2853E-005, v = 0.5200
calc: A = 3.1034E-006, v = 0.6136 (R2 = 0.9993); dv = +0.0936

Bas_3

exp: A = 2.2030E-006, v = 0.6400
calc: A = 1.5553E-006, v = 0.6635 (R2 = 0.9967); dv = +0.0235

Bas_4

exp: A = 6.1379E-007, v = 0.7000
calc: A = 2.2817E-006, v = 0.6120 (R2 = 0.9787); dv = -0.0880

Bas_1 - Vieille Coefficients

Bas_1

exp: A = 9.0004E-007, v = 0.7000
calc: A = 2.8234E-006, v = 0.6231 (R2 = 0.9989); dv = -0.0769

Bas_0 - Vieille Coefficients

Bas_0

exp: A = 1.7206E-006, v = 0.5900
calc: A = 1.5553E-006, v = 0.5968 (R2 = 0.9990); dv = +0.0068

Flame Structure & Heat-Flux Decomposition

q = total heat feedback to the surface (MW/m2). It splits exactly into three competing-flame pathways whose shares sum to 100%: out-kin = out-of-skeleton kinetic flame; skeleton = surface-weighted (metal burning + skeleton kinetic flame in the pores); diff = diffusion flame. h = flame heights (um). Ts_p / Ts_i = pocket / inter-pocket surface temperature (the two-temperature surface). Mean shares are the pressure-average per fuel. Numbers below reflect this report's input vector.

Bas_2 (includes Bas_3, Bas_4) - Flame Structure

Bas_2

p 1 MPa: q = 0.70 MW/m2; out-kin 72.9%, skeleton 6.0%, diff 21.1%
h[um] skel-kin 725.7, out-kin 262.9, diff 2025.8; Ts_p 613 K, Ts_i 621 K
p 1.611 MPa: q = 0.98 MW/m2; out-kin 79.4%, skeleton 8.5%, diff 12.1%
h[um] skel-kin 285.7, out-kin 178.2, diff 2515.4; Ts_p 617 K, Ts_i 631 K
p 2.222 MPa: q = 1.25 MW/m2; out-kin 81.1%, skeleton 10.8%, diff 8.1%
h[um] skel-kin 154.7, out-kin 139.2, diff 2955.6; Ts_p 621 K, Ts_i 638 K

p 2.833 MPa: q = 1.51 MW/m2; out-kin 81.2%, skeleton 12.9%, diff 5.9%
h[um] skel-kin 98.0, out-kin 116.3, diff 3362.3; Ts_p 623 K, Ts_i 644 K
p 3.444 MPa: q = 1.77 MW/m2; out-kin 80.5%, skeleton 15.0%, diff 4.5%
h[um] skel-kin 68.1, out-kin 101.1, diff 3744.6; Ts_p 626 K, Ts_i 649 K
p 4.056 MPa: q = 2.02 MW/m2; out-kin 79.6%, skeleton 16.9%, diff 3.6%
h[um] skel-kin 50.4, out-kin 90.1, diff 4108.6; Ts_p 628 K, Ts_i 653 K
p 4.667 MPa: q = 2.27 MW/m2; out-kin 78.4%, skeleton 18.7%, diff 2.9%
h[um] skel-kin 39.0, out-kin 81.8, diff 4458.3; Ts_p 630 K, Ts_i 656 K
p 5.278 MPa: q = 2.51 MW/m2; out-kin 77.2%, skeleton 20.4%, diff 2.5%
h[um] skel-kin 31.2, out-kin 75.3, diff 4796.4; Ts_p 631 K, Ts_i 659 K
p 5.889 MPa: q = 2.76 MW/m2; out-kin 75.9%, skeleton 22.0%, diff 2.1%
h[um] skel-kin 25.6, out-kin 69.9, diff 5125.2; Ts_p 633 K, Ts_i 662 K
p 6.5 MPa: q = 3.00 MW/m2; out-kin 74.6%, skeleton 23.6%, diff 1.8%
h[um] skel-kin 21.4, out-kin 65.6, diff 5446.1; Ts_p 634 K, Ts_i 665 K
mean shares: out-kin 78.1%, skeleton 15.5%, diffusion 6.5%

Bas_3

p 1 MPa: q = 1.23 MW/m2; out-kin 28.8%, skeleton 9.3%, diff 62.0%
h[um] skel-kin 459.8, out-kin 376.8, diff 401.1; Ts_p 621 K, Ts_i 621 K
p 1.611 MPa: q = 1.51 MW/m2; out-kin 38.7%, skeleton 17.3%, diff 44.0%
h[um] skel-kin 168.1, out-kin 236.5, diff 460.3; Ts_p 623 K, Ts_i 631 K
p 2.222 MPa: q = 1.81 MW/m2; out-kin 43.2%, skeleton 24.7%, diff 32.1%
h[um] skel-kin 88.4, out-kin 178.9, diff 523.5; Ts_p 626 K, Ts_i 638 K
p 2.833 MPa: q = 2.15 MW/m2; out-kin 44.8%, skeleton 31.1%, diff 24.1%
h[um] skel-kin 55.5, out-kin 147.8, diff 588.7; Ts_p 629 K, Ts_i 644 K
p 3.444 MPa: q = 2.50 MW/m2; out-kin 44.8%, skeleton 36.6%, diff 18.6%
h[um] skel-kin 38.6, out-kin 128.2, diff 654.7; Ts_p 631 K, Ts_i 649 K
p 4.056 MPa: q = 2.86 MW/m2; out-kin 44.0%, skeleton 41.3%, diff 14.8%
h[um] skel-kin 28.8, out-kin 114.7, diff 721.1; Ts_p 633 K, Ts_i 653 K
p 4.667 MPa: q = 3.23 MW/m2; out-kin 42.8%, skeleton 45.3%, diff 11.9%
h[um] skel-kin 22.4, out-kin 104.8, diff 787.8; Ts_p 635 K, Ts_i 656 K
p 5.278 MPa: q = 3.61 MW/m2; out-kin 41.4%, skeleton 48.8%, diff 9.8%
h[um] skel-kin 18.1, out-kin 97.2, diff 854.4; Ts_p 637 K, Ts_i 659 K
p 5.889 MPa: q = 4.00 MW/m2; out-kin 40.0%, skeleton 51.8%, diff 8.2%
h[um] skel-kin 15.0, out-kin 91.1, diff 921.1; Ts_p 638 K, Ts_i 662 K
p 6.5 MPa: q = 4.40 MW/m2; out-kin 38.5%, skeleton 54.5%, diff 7.0%
h[um] skel-kin 12.7, out-kin 86.2, diff 987.7; Ts_p 640 K, Ts_i 665 K
mean shares: out-kin 40.7%, skeleton 36.1%, diffusion 23.3%

Bas_4

p 1 MPa: q = 0.23 MW/m2; out-kin 7.0%, skeleton 24.0%, diff 69.0%
h[um] skel-kin 721.8, out-kin 1732.6, diff 1879.6; Ts_p 600 K, Ts_i 621 K
p 1.611 MPa: q = 0.28 MW/m2; out-kin 9.8%, skeleton 39.6%, diff 50.6%
h[um] skel-kin 255.4, out-kin 1050.3, diff 2094.5; Ts_p 602 K, Ts_i 631 K
p 2.222 MPa: q = 0.35 MW/m2; out-kin 10.6%, skeleton 53.7%, diff 35.8%
h[um] skel-kin 133.2, out-kin 786.4, diff 2373.2; Ts_p 605 K, Ts_i 638 K
p 2.833 MPa: q = 0.44 MW/m2; out-kin 10.3%, skeleton 64.3%, diff 25.3%
h[um] skel-kin 84.0, out-kin 651.1, diff 2693.2; Ts_p 607 K, Ts_i 644 K
p 3.444 MPa: q = 0.53 MW/m2; out-kin 9.7%, skeleton 72.0%, diff 18.4%
h[um] skel-kin 59.1, out-kin 569.4, diff 3039.5; Ts_p 610 K, Ts_i 649 K
p 4.056 MPa: q = 0.64 MW/m2; out-kin 8.9%, skeleton 77.4%, diff 13.6%
h[um] skel-kin 44.5, out-kin 514.7, diff 3403.0; Ts_p 612 K, Ts_i 653 K
p 4.667 MPa: q = 0.75 MW/m2; out-kin 8.2%, skeleton 81.4%, diff 10.4%
h[um] skel-kin 35.1, out-kin 475.1, diff 3778.5; Ts_p 614 K, Ts_i 656 K
p 5.278 MPa: q = 0.88 MW/m2; out-kin 7.5%, skeleton 84.4%, diff 8.1%
h[um] skel-kin 28.7, out-kin 445.0, diff 4162.3; Ts_p 616 K, Ts_i 659 K
p 5.889 MPa: q = 1.01 MW/m2; out-kin 6.9%, skeleton 86.6%, diff 6.5%
h[um] skel-kin 24.0, out-kin 421.1, diff 4552.4; Ts_p 618 K, Ts_i 662 K
p 6.5 MPa: q = 1.14 MW/m2; out-kin 6.4%, skeleton 88.4%, diff 5.2%
h[um] skel-kin 20.6, out-kin 401.6, diff 4947.4; Ts_p 620 K, Ts_i 665 K
mean shares: out-kin 8.5%, skeleton 67.2%, diffusion 24.3%

Bas_1 - Flame Structure

Bas_1

p 1 MPa: q = 0.70 MW/m2; out-kin 69.9%, skeleton 9.4%, diff 20.6%
h[um] skel-kin 836.1, out-kin 275.7, diff 2121.2; Ts_p 611 K, Ts_i 618 K
p 1.611 MPa: q = 0.97 MW/m2; out-kin 76.6%, skeleton 11.5%, diff 11.9%
h[um] skel-kin 305.6, out-kin 186.6, diff 2629.2; Ts_p 615 K, Ts_i 628 K
p 2.222 MPa: q = 1.24 MW/m2; out-kin 78.3%, skeleton 13.8%, diff 8.0%
h[um] skel-kin 157.2, out-kin 145.9, diff 3090.6; Ts_p 619 K, Ts_i 635 K
p 2.833 MPa: q = 1.50 MW/m2; out-kin 78.1%, skeleton 16.1%, diff 5.8%
h[um] skel-kin 95.7, out-kin 122.1, diff 3520.9; Ts_p 621 K, Ts_i 640 K
p 3.444 MPa: q = 1.76 MW/m2; out-kin 77.1%, skeleton 18.5%, diff 4.4%
h[um] skel-kin 64.5, out-kin 106.3, diff 3929.4; Ts_p 623 K, Ts_i 644 K
p 4.056 MPa: q = 2.02 MW/m2; out-kin 75.7%, skeleton 20.8%, diff 3.5%
h[um] skel-kin 46.5, out-kin 95.0, diff 4322.3; Ts_p 625 K, Ts_i 648 K
p 4.667 MPa: q = 2.27 MW/m2; out-kin 74.1%, skeleton 23.0%, diff 2.8%
h[um] skel-kin 35.2, out-kin 86.5, diff 4703.4; Ts_p 627 K, Ts_i 652 K
p 5.278 MPa: q = 2.53 MW/m2; out-kin 72.5%, skeleton 25.2%, diff 2.4%
h[um] skel-kin 27.6, out-kin 79.8, diff 5075.7; Ts_p 629 K, Ts_i 654 K
p 5.889 MPa: q = 2.79 MW/m2; out-kin 70.7%, skeleton 27.3%, diff 2.0%
h[um] skel-kin 22.2, out-kin 74.4, diff 5441.2; Ts_p 630 K, Ts_i 657 K
p 6.5 MPa: q = 3.05 MW/m2; out-kin 69.0%, skeleton 29.3%, diff 1.7%
h[um] skel-kin 18.3, out-kin 70.0, diff 5801.4; Ts_p 631 K, Ts_i 660 K
mean shares: out-kin 74.2%, skeleton 19.5%, diffusion 6.3%

Bas_0 - Flame Structure

Bas_0

p 1 MPa: q = 2.02 MW/m2; out-kin 36.0%, skeleton 46.7%, diff 17.3%
h[um] skel-kin 48.0, out-kin 87.6, diff 812.6; Ts_p 830 K, Ts_i 838 K
p 1.611 MPa: q = 2.47 MW/m2; out-kin 52.6%, skeleton 35.7%, diff 11.8%
h[um] skel-kin 44.9, out-kin 57.5, diff 975.6; Ts_p 834 K, Ts_i 853 K
p 2.222 MPa: q = 2.90 MW/m2; out-kin 63.2%, skeleton 28.2%, diff 8.6%
h[um] skel-kin 44.0, out-kin 44.3, diff 1131.1; Ts_p 838 K, Ts_i 864 K
p 2.833 MPa: q = 3.32 MW/m2; out-kin 70.2%, skeleton 23.1%, diff 6.7%
h[um] skel-kin 43.7, out-kin 36.8, diff 1279.1; Ts_p 841 K, Ts_i 872 K
p 3.444 MPa: q = 3.73 MW/m2; out-kin 75.2%, skeleton 19.4%, diff 5.3%
h[um] skel-kin 43.8, out-kin 31.9, diff 1420.3; Ts_p 844 K, Ts_i 878 K
p 4.056 MPa: q = 4.12 MW/m2; out-kin 78.9%, skeleton 16.7%, diff 4.4%
h[um] skel-kin 43.9, out-kin 28.4, diff 1555.5; Ts_p 846 K, Ts_i 884 K
p 4.667 MPa: q = 4.50 MW/m2; out-kin 81.6%, skeleton 14.6%, diff 3.7%
h[um] skel-kin 44.2, out-kin 25.7, diff 1685.3; Ts_p 849 K, Ts_i 888 K
p 5.278 MPa: q = 4.86 MW/m2; out-kin 83.8%, skeleton 13.0%, diff 3.2%
h[um] skel-kin 44.4, out-kin 23.6, diff 1810.3; Ts_p 850 K, Ts_i 893 K
p 5.889 MPa: q = 5.22 MW/m2; out-kin 85.5%, skeleton 11.7%, diff 2.8%
h[um] skel-kin 44.7, out-kin 21.9, diff 1931.2; Ts_p 852 K, Ts_i 897 K
p 6.5 MPa: q = 5.57 MW/m2; out-kin 86.9%, skeleton 10.6%, diff 2.5%
h[um] skel-kin 45.0, out-kin 20.5, diff 2048.3; Ts_p 854 K, Ts_i 900 K
mean shares: out-kin 71.4%, skeleton 22.0%, diffusion 6.6%

Propellant Composition "Bas_2"

At pressure 1 MPa

Experimental burning rate 16.94 mm/s
Calculated burning rate 15.15 mm/s

First Conditional Fuel

Surface temperature 620.57 K
Linear burning rate 11.58 mm/s
Mass decomposition rate 19.22 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 1.23e+06 W/m^2

Kinetic flame height 251.2 μm
Average kinetic flame density 1.83 kg/m^3
Average kinetic flame temperature 2,163.78 K
Binder sublimation heat flux density 1.23e+06 W/m^2
Surface heat balance error 6.4e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 612.98 K
Linear burning rate 8.03 mm/s
Mass decomposition rate 13.31 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 666,404.95 W/m^2
Kinetic flame height 262.91 μm
Average kinetic flame density 2.67 kg/m^3
Average kinetic flame temperature 1,488.99 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 134,764.9 W/m^2
Kinetic flame height 725.72 μm
Average kinetic flame density 3.6 kg/m^3
Average kinetic flame temperature 1,101.99 K
Heat flux density due to metal combustion, 44,657.46 W/m^2
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0073 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.88e-05 m
Pore diameter 4.54e-06 m

Diffusion flame heat flux density 147,643.25 W/m^2
Diffusion flame height 2,025.84 μm
Heat flux density to surface within skeletal structure 42,108.44 W/m^2
Heat flux density to surface outside skeletal structure 510,007.1 W/m^2
Total heat flux density to surface 699,758.78 W/m^2
Binder sublimation heat flux density 699,758.78 W/m^2
Surface heat balance error -3.79e-04 W/m^2

At pressure 1.61 MPa

Experimental burning rate 21.71 mm/s
Calculated burning rate 19.89 mm/s

First Conditional Fuel

Surface temperature 630.9 K
Linear burning rate 18.84 mm/s
Mass decomposition rate 31.25 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 2.48e+06 W/m^2
Kinetic flame height 123.91 μm
Average kinetic flame density 2.95 kg/m^3
Average kinetic flame temperature 2,168.95 K
Binder sublimation heat flux density 2.48e+06 W/m^2
Surface heat balance error -5.78e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 617.43 K
Linear burning rate 9.96 mm/s
Mass decomposition rate 16.53 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 980,450.85 W/m^2
Kinetic flame height 178.24 μm
Average kinetic flame density 4.29 kg/m^3
Average kinetic flame temperature 1,491.22 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 340,712.96 W/m^2
Kinetic flame height 285.74 μm

Average kinetic flame density 5.79 kg/m^3
Average kinetic flame temperature 1,104.22 K
Heat flux density due to metal combustion, 60,364.74 W/m^2
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0064 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.86e-05 m
Pore diameter 4e-06 m
Diffusion flame heat flux density 118,733.63 W/m^2
Diffusion flame height 2,515.35 μm
Heat flux density to surface within skeletal structure 83,119.99 W/m^2
Heat flux density to surface outside skeletal structure 777,260.64 W/m^2
Total heat flux density to surface 979,114.25 W/m^2
Binder sublimation heat flux density 979,114.25 W/m^2
Surface heat balance error -5.15e-04 W/m^2

At pressure 2.22 MPa
Experimental burning rate 25.67 mm/s
Calculated burning rate 24.09 mm/s
First Conditional Fuel
Surface temperature 638.25 K
Linear burning rate 26.36 mm/s
Mass decomposition rate 43.74 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 3.96e+06 W/m^2
Kinetic flame height 77.57 μm
Average kinetic flame density 4.06 kg/m^3
Average kinetic flame temperature 2,172.62 K
Binder sublimation heat flux density 3.96e+06 W/m^2
Surface heat balance error 0.0014 W/m^2
Secondary Conditional Fuel
Surface temperature 620.79 K
Linear burning rate 11.71 mm/s
Mass decomposition rate 19.42 kg·s^{-1}·m^{-2}
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.25e+06 W/m^2
Kinetic flame height 139.23 μm
Average kinetic flame density 5.91 kg/m^3
Average kinetic flame temperature 1,492.89 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 627,143.09 W/m^2
Kinetic flame height 154.7 μm
Average kinetic flame density 7.98 kg/m^3
Average kinetic flame temperature 1,105.89 K
Heat flux density due to metal combustion, 75,577.31 W/m^2
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0059 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.27e-05 m
Pore diameter 3.64e-06 m
Diffusion flame heat flux density 100,933.14 W/m^2
Diffusion flame height 2,955.63 μm
Heat flux density to surface within skeletal structure 134,671.94 W/m^2
Heat flux density to surface outside skeletal structure 1.01e+06 W/m^2
Total heat flux density to surface 1.25e+06 W/m^2
Binder sublimation heat flux density 1.25e+06 W/m^2
Surface heat balance error -7.66e-05 W/m^2

<u>At pressure 2.83 MPa</u>
Experimental burning rate 29.12 mm/s
Calculated burning rate 27.94 mm/s
First Conditional Fuel
Surface temperature 643.99 K
Linear burning rate 34.1 mm/s
Mass decomposition rate 56.58 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 5.61e+06 W/m^2
Kinetic flame height 54.65 μm
Average kinetic flame density 5.17 kg/m^3
Average kinetic flame temperature 2,175.5 K
Binder sublimation heat flux density 5.61e+06 W/m^2
Surface heat balance error -0.0025 W/m^2
Secondary Conditional Fuel
Surface temperature 623.5 K
Linear burning rate 13.32 mm/s
Mass decomposition rate 22.1 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.5e+06 W/m^2
Kinetic flame height 116.32 μm
Average kinetic flame density 7.53 kg/m^3
Average kinetic flame temperature 1,494.25 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 987,430.98 W/m^2
Kinetic flame height 97.98 μm
Average kinetic flame density 10.16 kg/m^3
Average kinetic flame temperature 1,107.25 K
Heat flux density due to metal combustion, 90,453.42 W/m^2
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0054 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.9e-05 m
Pore diameter 3.37e-06 m
Diffusion flame heat flux density 88,645.96 W/m^2
Diffusion flame height 3,362.26 μm
Heat flux density to surface within skeletal structure 195,281.53 W/m^2
Heat flux density to surface outside skeletal structure 1.23e+06 W/m^2
Total heat flux density to surface 1.51e+06 W/m^2
Binder sublimation heat flux density 1.51e+06 W/m^2
Surface heat balance error -3.48e-04 W/m^2

<u>At pressure 3.44 MPa</u>
Experimental burning rate 32.23 mm/s
Calculated burning rate 31.55 mm/s
First Conditional Fuel
Surface temperature 648.73 K
Linear burning rate 42.03 mm/s
Mass decomposition rate 69.72 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 7.4e+06 W/m^2
Kinetic flame height 41.31 μm
Average kinetic flame density 6.28 kg/m^3
Average kinetic flame temperature 2,177.86 K
Binder sublimation heat flux density 7.4e+06 W/m^2
Surface heat balance error -0.0026 W/m^2
Secondary Conditional Fuel
Surface temperature 625.78 K
Linear burning rate 14.83 mm/s
Mass decomposition rate 24.61 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.72e+06 W/m²
Kinetic flame height 101.08 µm
Average kinetic flame density 9.14 kg/m³
Average kinetic flame temperature 1,495.39 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.42e+06 W/m²
Kinetic flame height 68.14 µm
Average kinetic flame density 12.33 kg/m³
Average kinetic flame temperature 1,108.39 K
Heat flux density due to metal combustion, 105,110.06 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0051 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.63e-05 m
Pore diameter 3.16e-06 m

Diffusion flame heat flux density 79,534.04 W/m²
Diffusion flame height 3,744.59 µm
Heat flux density to surface within skeletal structure 263,985.49 W/m²
Heat flux density to surface outside skeletal structure 1.42e+06 W/m²
Total heat flux density to surface 1.77e+06 W/m²
Binder sublimation heat flux density 1.77e+06 W/m²
Surface heat balance error -4.67e-04 W/m²

<u>*At pressure 4.06 MPa*</u>

Experimental burning rate 35.09 mm/s
Calculated burning rate 34.99 mm/s

First Conditional Fuel

Surface temperature 652.77 K
Linear burning rate 50.11 mm/s
Mass decomposition rate 83.12 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 9.33e+06 W/m²
Kinetic flame height 32.74 µm
Average kinetic flame density 7.38 kg/m³
Average kinetic flame temperature 2,179.89 K
Binder sublimation heat flux density 9.33e+06 W/m²
Surface heat balance error 0.0036 W/m²

Secondary Conditional Fuel

Surface temperature 627.76 K
Linear burning rate 16.28 mm/s
Mass decomposition rate 27 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.93e+06 W/m²
Kinetic flame height 90.11 µm
Average kinetic flame density 10.76 kg/m³
Average kinetic flame temperature 1,496.38 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.91e+06 W/m²
Kinetic flame height 50.42 µm
Average kinetic flame density 14.51 kg/m³
Average kinetic flame temperature 1,109.38 K
Heat flux density due to metal combustion, 119,628.29 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0048 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.43e-05 m
Pore diameter 2.99e-06 m

Diffusion flame heat flux density 72,439.64 W/m²
Diffusion flame height 4,108.58 μm
Heat flux density to surface within skeletal structure 340,068.26 W/m²
Heat flux density to surface outside skeletal structure 1.6e+06 W/m²
Total heat flux density to surface 2.02e+06 W/m²
Binder sublimation heat flux density 2.02e+06 W/m²
Surface heat balance error -2.27e-04 W/m²

At pressure 4.67 MPa
Experimental burning rate 37.75 mm/s
Calculated burning rate 38.28 mm/s
First Conditional Fuel
Surface temperature 656.3 K
Linear burning rate 58.33 mm/s
Mass decomposition rate 96.76 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.14e+07 W/m²
Kinetic flame height 26.82 μm
Average kinetic flame density 8.49 kg/m³
Average kinetic flame temperature 2,181.65 K
Binder sublimation heat flux density 1.14e+07 W/m²
Surface heat balance error -0.003 W/m²
Secondary Conditional Fuel
Surface temperature 629.51 K
Linear burning rate 17.66 mm/s
Mass decomposition rate 29.3 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.12e+06 W/m²
Kinetic flame height 81.8 μm
Average kinetic flame density 12.37 kg/m³
Average kinetic flame temperature 1,497.25 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 2.47e+06 W/m²
Kinetic flame height 39 μm
Average kinetic flame density 16.68 kg/m³
Average kinetic flame temperature 1,110.25 K
Heat flux density due to metal combustion, 134,065.34 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0046 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.27e-05 m
Pore diameter 2.85e-06 m
Diffusion flame heat flux density 66,718.77 W/m²
Diffusion flame height 4,458.25 μm
Heat flux density to surface within skeletal structure 422,964.71 W/m²
Heat flux density to surface outside skeletal structure 1.78e+06 W/m²
Total heat flux density to surface 2.27e+06 W/m²
Binder sublimation heat flux density 2.27e+06 W/m²
Surface heat balance error -8.52e-04 W/m²

At pressure 5.28 MPa
Experimental burning rate 40.24 mm/s
Calculated burning rate 41.47 mm/s
First Conditional Fuel
Surface temperature 659.45 K
Linear burning rate 66.67 mm/s
Mass decomposition rate 110.61 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.35e+07 W/m²
Kinetic flame height 22.54 μm

Average kinetic flame density 9.59 kg/m³
Average kinetic flame temperature 2,183.22 K
Binder sublimation heat flux density 1.35e+07 W/m²
Surface heat balance error 0.003 W/m²

Secondary Conditional Fuel

Surface temperature 631.09 K
Linear burning rate 19 mm/s
Mass decomposition rate 31.52 $kg·s^{-1}·m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.3e+06 W/m²
Kinetic flame height 75.25 μm
Average kinetic flame density 13.98 kg/m³
Average kinetic flame temperature 1,498.04 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.08e+06 W/m²
Kinetic flame height 31.18 μm
Average kinetic flame density 18.85 kg/m³
Average kinetic flame temperature 1,111.04 K
Heat flux density due to metal combustion, 148,462.07 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0044 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.15e-05 m
Pore diameter 2.73e-06 m

Diffusion flame heat flux density 61,982.09 W/m²
Diffusion flame height 4,796.41 μm
Heat flux density to surface within skeletal structure 512,132.05 W/m²
Heat flux density to surface outside skeletal structure 1.94e+06 W/m²
Total heat flux density to surface 2.51e+06 W/m²
Binder sublimation heat flux density 2.51e+06 W/m²
Surface heat balance error 0.0012 W/m²

At pressure 5.89 MPa

Experimental burning rate 42.6 mm/s
Calculated burning rate 44.56 mm/s

First Conditional Fuel

Surface temperature 662.28 K
Linear burning rate 75.13 mm/s
Mass decomposition rate 124.65 $kg·s^{-1}·m^{-2}$
Kinetic flame heat flux density to surface 1.58e+07 W/m²
Kinetic flame height 19.31 μm
Average kinetic flame density 10.7 kg/m³
Average kinetic flame temperature 2,184.64 K
Binder sublimation heat flux density 1.58e+07 W/m²
Surface heat balance error 0.0041 W/m²

Secondary Conditional Fuel

Surface temperature 632.52 K
Linear burning rate 20.3 mm/s
Mass decomposition rate 33.68 $kg·s^{-1}·m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2.48e+06 W/m²
Kinetic flame height 69.95 μm
Average kinetic flame density 15.59 kg/m³
Average kinetic flame temperature 1,498.76 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.75e+06 W/m²
Kinetic flame height 25.58 μm
Average kinetic flame density 21.02 kg/m³

Average kinetic flame temperature 1,111.76 K
Heat flux density due to metal combustion, 162,851.86 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0042 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.04e-05 m
Pore diameter 2.63e-06 m
Diffusion flame heat flux density 57,977.95 W/m²
Diffusion flame height 5,125.19 μm
Heat flux density to surface within skeletal structure 607,406.36 W/m²
Heat flux density to surface outside skeletal structure 2.09e+06 W/m²
Total heat flux density to surface 2.76e+06 W/m²
Binder sublimation heat flux density 2.76e+06 W/m²
Surface heat balance error -8.69e-04 W/m²

At pressure 6.5 MPa
Experimental burning rate 44.85 mm/s
Calculated burning rate 47.58 mm/s
First Conditional Fuel
Surface temperature 664.86 K
Linear burning rate 83.7 mm/s
Mass decomposition rate 138.86 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.81e+07 W/m²
Kinetic flame height 16.8 μm
Average kinetic flame density 11.8 kg/m³
Average kinetic flame temperature 2,185.93 K
Binder sublimation heat flux density 1.81e+07 W/m²
Surface heat balance error -7.73e-05 W/m²
Secondary Conditional Fuel
Surface temperature 633.84 K
Linear burning rate 21.57 mm/s
Mass decomposition rate 35.79 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.64e+06 W/m²
Kinetic flame height 65.55 μm
Average kinetic flame density 17.21 kg/m³
Average kinetic flame temperature 1,499.42 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 4.47e+06 W/m²
Kinetic flame height 21.42 μm
Average kinetic flame density 23.19 kg/m³
Average kinetic flame temperature 1,112.42 K
Heat flux density due to metal combustion, 177,256.07 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.0041 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 9.58e-06 m
Pore diameter 2.54e-06 m
Diffusion flame heat flux density 54,537.29 W/m²
Diffusion flame height 5,446.1 μm
Heat flux density to surface within skeletal structure 708,214.41 W/m²
Heat flux density to surface outside skeletal structure 2.24e+06 W/m²
Total heat flux density to surface 3e+06 W/m²
Binder sublimation heat flux density 3e+06 W/m²
Surface heat balance error 7.87e-04 W/m²

Propellant Composition "Bas_3"

At pressure 1 MPa

Experimental burning rate 15.24 mm/s

Calculated burning rate 15.47 mm/s

First Conditional Fuel

Surface temperature 620.57 K

Linear burning rate 11.58 mm/s

Mass decomposition rate 19.22 $kg \cdot s^{-1} \cdot m^{-2}$

Kinetic flame heat flux density to surface 1.23e+06 W/m^2

Kinetic flame height 251.2 μm

Average kinetic flame density 1.83 kg/m^3

Average kinetic flame temperature 2,163.78 K

Binder sublimation heat flux density 1.23e+06 W/m^2

Surface heat balance error 6.4e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 620.57 K

Linear burning rate 11.58 mm/s

Mass decomposition rate 19.22 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 462,901.24 W/m^2

Kinetic flame height 376.85 μm

Average kinetic flame density 2.66 kg/m^3

Average kinetic flame temperature 1,492.78 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 435,508.41 W/m^2

Kinetic flame height 459.79 μm

Average kinetic flame density 2.45 kg/m^3

Average kinetic flame temperature 1,621.78 K

Heat flux density due to metal combustion, 46,915.98 W/m^2

Effective thermal conductivity 0.26 W/(m·K)

Conductive thermal conductivity 0.22 W/(m·K)

Radiative thermal conductivity 0.036 W/(m·K)

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness 2.31e-05 m

Pore diameter 3.66e-06 m

Diffusion flame heat flux density 761,170.48 W/m^2

Diffusion flame height 401.15 μm

Heat flux density to surface within skeletal structure 113,996.21 W/m^2

Heat flux density to surface outside skeletal structure 353,518.32 W/m^2

Total heat flux density to surface 1.23e+06 W/m^2

Binder sublimation heat flux density 1.23e+06 W/m^2

Surface heat balance error -1.47e-04 W/m^2

At pressure 1.61 MPa

Experimental burning rate 20.68 mm/s

Calculated burning rate 20.32 mm/s

First Conditional Fuel

Surface temperature 630.9 K

Linear burning rate 18.84 mm/s

Mass decomposition rate 31.25 $kg \cdot s^{-1} \cdot m^{-2}$

Kinetic flame heat flux density to surface 2.48e+06 W/m^2

Kinetic flame height 123.91 μm

Average kinetic flame density 2.95 kg/m^3

Average kinetic flame temperature 2,168.95 K

Binder sublimation heat flux density 2.48e+06 W/m^2

Surface heat balance error -5.78e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 623.45 K

Linear burning rate 13.29 mm/s

Mass decomposition rate 22.05 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 736,509.9 W/m^2
Kinetic flame height 236.46 μm
Average kinetic flame density 4.28 kg/m^3
Average kinetic flame temperature 1,494.23 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.19e+06 W/m^2
Kinetic flame height 168.14 μm
Average kinetic flame density 3.94 kg/m^3
Average kinetic flame temperature 1,623.23 K
Heat flux density due to metal combustion, 56,273.1 W/m^2
Effective thermal conductivity 0.26 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.033 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.9e-05 m
Pore diameter 3.37e-06 m
Diffusion flame heat flux density 662,723.79 W/m^2
Diffusion flame height 460.3 μm
Heat flux density to surface within skeletal structure 259,908.44 W/m^2
Heat flux density to surface outside skeletal structure 582,810.67 W/m^2
Total heat flux density to surface 1.51e+06 W/m^2
Binder sublimation heat flux density 1.51e+06 W/m^2
Surface heat balance error -7.01e-04 W/m^2

At pressure 2.22 MPa
Experimental burning rate 25.41 mm/s
Calculated burning rate 24.72 mm/s
First Conditional Fuel
Surface temperature 638.25 K
Linear burning rate 26.36 mm/s
Mass decomposition rate 43.74 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 3.96e+06 W/m^2
Kinetic flame height 77.57 μm
Average kinetic flame density 4.06 kg/m^3
Average kinetic flame temperature 2,172.62 K
Binder sublimation heat flux density 3.96e+06 W/m^2
Surface heat balance error 0.0014 W/m^2
Secondary Conditional Fuel
Surface temperature 626.18 K
Linear burning rate 15.12 mm/s
Mass decomposition rate 25.08 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 971,993.16 W/m^2
Kinetic flame height 178.89 μm
Average kinetic flame density 5.9 kg/m^3
Average kinetic flame temperature 1,495.59 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 2.26e+06 W/m^2
Kinetic flame height 88.43 μm
Average kinetic flame density 5.43 kg/m^3
Average kinetic flame temperature 1,624.59 K
Heat flux density due to metal combustion, 66,758.34 W/m^2
Effective thermal conductivity 0.25 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.031 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.58e-05 m

Pore diameter 3.13e-06 m
Diffusion flame heat flux density 582,150.07 W/m²
Diffusion flame height 523.54 μm
Heat flux density to surface within skeletal structure 448,157.21 W/m²
Heat flux density to surface outside skeletal structure 784,624.21 W/m²
Total heat flux density to surface 1.81e+06 W/m²
Binder sublimation heat flux density 1.81e+06 W/m²
Surface heat balance error -8.98e-04 W/m²

At pressure 2.83 MPa
Experimental burning rate 29.68 mm/s
Calculated burning rate 28.93 mm/s
First Conditional Fuel
Surface temperature 643.99 K
Linear burning rate 34.1 mm/s
Mass decomposition rate 56.58 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 5.61e+06 W/m²
Kinetic flame height 54.65 μm
Average kinetic flame density 5.17 kg/m³
Average kinetic flame temperature 2,175.5 K
Binder sublimation heat flux density 5.61e+06 W/m²
Surface heat balance error -0.0025 W/m²
Secondary Conditional Fuel
Surface temperature 628.69 K
Linear burning rate 17 mm/s
Mass decomposition rate 28.2 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.18e+06 W/m²
Kinetic flame height 147.76 μm
Average kinetic flame density 7.51 kg/m³
Average kinetic flame temperature 1,496.84 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 3.59e+06 W/m²
Kinetic flame height 55.51 μm
Average kinetic flame density 6.92 kg/m³
Average kinetic flame temperature 1,625.84 K
Heat flux density due to metal combustion, 78,039.45 W/m²
Effective thermal conductivity 0.25 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.029 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.34e-05 m
Pore diameter 2.92e-06 m
Diffusion flame heat flux density 517,320.44 W/m²
Diffusion flame height 588.67 μm
Heat flux density to surface within skeletal structure 668,193.6 W/m²
Heat flux density to surface outside skeletal structure 961,188.11 W/m²
Total heat flux density to surface 2.15e+06 W/m²
Binder sublimation heat flux density 2.15e+06 W/m²
Surface heat balance error -3.09e-04 W/m²

At pressure 3.44 MPa
Experimental burning rate 33.63 mm/s
Calculated burning rate 33.06 mm/s
First Conditional Fuel
Surface temperature 648.73 K
Linear burning rate 42.03 mm/s
Mass decomposition rate 69.72 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 7.4e+06 W/m²

Kinetic flame height 41.31 μm
Average kinetic flame density 6.28 kg/m^3
Average kinetic flame temperature 2,177.86 K
Binder sublimation heat flux density 7.4e+06 W/m^2
Surface heat balance error -0.0026 W/m^2

Secondary Conditional Fuel

Surface temperature 630.98 K
Linear burning rate 18.91 mm/s
Mass decomposition rate 31.37 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.35e+06 W/m^2
Kinetic flame height 128.21 μm
Average kinetic flame density 9.13 kg/m^3
Average kinetic flame temperature 1,497.99 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 5.16e+06 W/m^2
Kinetic flame height 38.63 μm
Average kinetic flame density 8.4 kg/m^3
Average kinetic flame temperature 1,626.99 K
Heat flux density due to metal combustion, 89,943.7 W/m^2
Effective thermal conductivity 0.25 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.027 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.15e-05 m
Pore diameter 2.74e-06 m

Diffusion flame heat flux density 464,802.22 W/m^2
Diffusion flame height 654.69 μm
Heat flux density to surface within skeletal structure 913,373.38 W/m^2
Heat flux density to surface outside skeletal structure 1.12e+06 W/m^2
Total heat flux density to surface 2.5e+06 W/m^2
Binder sublimation heat flux density 2.5e+06 W/m^2
Surface heat balance error 3.51e-04 W/m^2

<u>*At pressure 4.06 MPa*</u>

Experimental burning rate 37.34 mm/s
Calculated burning rate 37.13 mm/s

First Conditional Fuel

Surface temperature 652.77 K
Linear burning rate 50.11 mm/s
Mass decomposition rate 83.12 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 9.33e+06 W/m^2
Kinetic flame height 32.74 μm
Average kinetic flame density 7.38 kg/m^3
Average kinetic flame temperature 2,179.89 K
Binder sublimation heat flux density 9.33e+06 W/m^2
Surface heat balance error 0.0036 W/m^2

Secondary Conditional Fuel

Surface temperature 633.07 K
Linear burning rate 20.83 mm/s
Mass decomposition rate 34.55 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.51e+06 W/m^2
Kinetic flame height 114.74 μm
Average kinetic flame density 10.74 kg/m^3
Average kinetic flame temperature 1,499.04 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 6.92e+06 W/m^2
Kinetic flame height 28.75 μm

Average kinetic flame density 9.89 kg/m³
Average kinetic flame temperature 1,628.04 K
Heat flux density due to metal combustion, 102,371.22 W/m²
Effective thermal conductivity 0.25 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.026 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 1.01e-05 m
Pore diameter 2.59e-06 m

Diffusion flame heat flux density 421,685.64 W/m²
Diffusion flame height 721.14 μm
Heat flux density to surface within skeletal structure 1.18e+06 W/m²
Heat flux density to surface outside skeletal structure 1.26e+06 W/m²
Total heat flux density to surface 2.86e+06 W/m²
Binder sublimation heat flux density 2.86e+06 W/m²
Surface heat balance error -5.74e-04 W/m²

At pressure 4.67 MPa

Experimental burning rate 40.85 mm/s
Calculated burning rate 41.17 mm/s

First Conditional Fuel

Surface temperature 656.3 K
Linear burning rate 58.33 mm/s
Mass decomposition rate 96.76 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.14e+07 W/m²
Kinetic flame height 26.82 μm
Average kinetic flame density 8.49 kg/m³
Average kinetic flame temperature 2,181.65 K
Binder sublimation heat flux density 1.14e+07 W/m²
Surface heat balance error -0.003 W/m²

Secondary Conditional Fuel

Surface temperature 635 K
Linear burning rate 22.75 mm/s
Mass decomposition rate 37.74 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.65e+06 W/m²
Kinetic flame height 104.83 μm
Average kinetic flame density 12.35 kg/m³
Average kinetic flame temperature 1,500 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 8.87e+06 W/m²
Kinetic flame height 22.42 μm
Average kinetic flame density 11.37 kg/m³
Average kinetic flame temperature 1,629 K
Heat flux density due to metal combustion, 115,257.4 W/m²
Effective thermal conductivity 0.25 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.024 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 8.89e-06 m
Pore diameter 2.46e-06 m

Diffusion flame heat flux density 385,778.97 W/m²
Diffusion flame height 787.76 μm
Heat flux density to surface within skeletal structure 1.46e+06 W/m²
Heat flux density to surface outside skeletal structure 1.38e+06 W/m²
Total heat flux density to surface 3.23e+06 W/m²
Binder sublimation heat flux density 3.23e+06 W/m²
Surface heat balance error -1.14e-04 W/m²

At pressure 5.28 MPa

Experimental burning rate 44.2 mm/s

Calculated burning rate 45.18 mm/s

First Conditional Fuel

Surface temperature 659.45 K

Linear burning rate 66.67 mm/s

Mass decomposition rate 110.61 $kg\cdot s^{-1}\cdot m^{-2}$

Kinetic flame heat flux density to surface 1.35e+07 W/m^2

Kinetic flame height 22.54 μm

Average kinetic flame density 9.59 kg/m^3

Average kinetic flame temperature 2,183.22 K

Binder sublimation heat flux density 1.35e+07 W/m^2

Surface heat balance error 0.003 W/m^2

Secondary Conditional Fuel

Surface temperature 636.79 K

Linear burning rate 24.67 mm/s

Mass decomposition rate 40.94 $kg\cdot s^{-1}\cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.78e+06 W/m^2

Kinetic flame height 97.21 μm

Average kinetic flame density 13.96 kg/m^3

Average kinetic flame temperature 1,500.89 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.1e+07 W/m^2

Kinetic flame height 18.1 μm

Average kinetic flame density 12.85 kg/m^3

Average kinetic flame temperature 1,629.89 K

Heat flux density due to metal combustion, 128,558.9 W/m^2

Effective thermal conductivity 0.24 W/(m·K)

Conductive thermal conductivity 0.22 W/(m·K)

Radiative thermal conductivity 0.023 W/(m·K)

Conductive thermal conductivity balance error -5.5152085010057306E-08

Skeleton layer thickness 7.92e-06 m

Pore diameter 2.34e-06 m

Diffusion flame heat flux density 355,467.27 W/m^2

Diffusion flame height 854.43 μm

Heat flux density to surface within skeletal structure 1.76e+06 W/m^2

Heat flux density to surface outside skeletal structure 1.5e+06 W/m^2

Total heat flux density to surface 3.61e+06 W/m^2

Binder sublimation heat flux density 3.61e+06 W/m^2

Surface heat balance error 1.88e-04 W/m^2

At pressure 5.89 MPa

Experimental burning rate 47.41 mm/s

Calculated burning rate 49.17 mm/s

First Conditional Fuel

Surface temperature 662.28 K

Linear burning rate 75.13 mm/s

Mass decomposition rate 124.65 $kg\cdot s^{-1}\cdot m^{-2}$

Kinetic flame heat flux density to surface 1.58e+07 W/m^2

Kinetic flame height 19.31 μm

Average kinetic flame density 10.7 kg/m^3

Average kinetic flame temperature 2,184.64 K

Binder sublimation heat flux density 1.58e+07 W/m^2

Surface heat balance error 0.0041 W/m^2

Secondary Conditional Fuel

Surface temperature 638.44 K

Linear burning rate 26.6 mm/s

Mass decomposition rate 44.13 $kg\cdot s^{-1}\cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 1.89e+06 W/m²
Kinetic flame height 91.15 μm
Average kinetic flame density 15.56 kg/m³
Average kinetic flame temperature 1,501.72 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.32e+07 W/m²
Kinetic flame height 15 μm
Average kinetic flame density 14.33 kg/m³
Average kinetic flame temperature 1,630.72 K
Heat flux density due to metal combustion, 142,243.45 W/m²
Effective thermal conductivity 0.24 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.022 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 7.12e-06 m
Pore diameter 2.24e-06 m

Diffusion flame heat flux density 329,563.38 W/m²
Diffusion flame height 921.08 μm
Heat flux density to surface within skeletal structure 2.07e+06 W/m²
Heat flux density to surface outside skeletal structure 1.6e+06 W/m²
Total heat flux density to surface 4e+06 W/m²
Binder sublimation heat flux density 4e+06 W/m²
Surface heat balance error 4.03e-04 W/m²

<u>*At pressure 6.5 MPa*</u>

Experimental burning rate 50.5 mm/s
Calculated burning rate 53.14 mm/s

First Conditional Fuel

Surface temperature 664.86 K
Linear burning rate 83.7 mm/s
Mass decomposition rate 138.86 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.81e+07 W/m²
Kinetic flame height 16.8 μm
Average kinetic flame density 11.8 kg/m³
Average kinetic flame temperature 2,185.93 K
Binder sublimation heat flux density 1.81e+07 W/m²
Surface heat balance error -7.73e-05 W/m²

Secondary Conditional Fuel

Surface temperature 639.99 K
Linear burning rate 28.52 mm/s
Mass decomposition rate 47.32 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 2e+06 W/m²
Kinetic flame height 86.18 μm
Average kinetic flame density 17.17 kg/m³
Average kinetic flame temperature 1,502.5 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.56e+07 W/m²
Kinetic flame height 12.69 μm
Average kinetic flame density 15.81 kg/m³
Average kinetic flame temperature 1,631.5 K
Heat flux density due to metal combustion, 156,287.22 W/m²
Effective thermal conductivity 0.24 W/(m·K)
Conductive thermal conductivity 0.22 W/(m·K)
Radiative thermal conductivity 0.021 W/(m·K)
Conductive thermal conductivity balance error -5.5152085010057306E-08
Skeleton layer thickness 6.45e-06 m
Pore diameter 2.15e-06 m

Diffusion flame heat flux density 307,181.98 W/m²
Diffusion flame height 987.69 μm
Heat flux density to surface within skeletal structure 2.4e+06 W/m²
Heat flux density to surface outside skeletal structure 1.7e+06 W/m²
Total heat flux density to surface 4.4e+06 W/m²
Binder sublimation heat flux density 4.4e+06 W/m²
Surface heat balance error 0.0017 W/m²

Propellant Composition "Bas_4"

At pressure 1 MPa
Experimental burning rate 9.73 mm/s
Calculated burning rate 11.79 mm/s
First Conditional Fuel
Surface temperature 620.57 K
Linear burning rate 11.58 mm/s
Mass decomposition rate 19.22 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 1.23e+06 W/m²
Kinetic flame height 251.2 μm
Average kinetic flame density 1.83 kg/m³
Average kinetic flame temperature 2,163.78 K
Binder sublimation heat flux density 1.23e+06 W/m²
Surface heat balance error 6.4e-04 W/m²
Secondary Conditional Fuel
Surface temperature 600 K
Linear burning rate 4.19 mm/s
Mass decomposition rate 6.95 kg·s⁻¹·m⁻²
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 25,510.21 W/m²
Kinetic flame height 1,732.64 μm
Average kinetic flame density 4.83 kg/m³
Average kinetic flame temperature 821 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 88,801.83 W/m²
Kinetic flame height 721.83 μm
Average kinetic flame density 4.31 kg/m³
Average kinetic flame temperature 920.5 K
Heat flux density due to metal combustion, 61,738.89 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.0041 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 9.74e-05 m
Pore diameter 6.66e-06 m
Diffusion flame heat flux density 158,543.34 W/m²
Diffusion flame height 1,879.61 μm
Heat flux density to surface within skeletal structure 55,165.88 W/m²
Heat flux density to surface outside skeletal structure 16,161.95 W/m²
Total heat flux density to surface 229,871.17 W/m²
Binder sublimation heat flux density 229,871.17 W/m²
Surface heat balance error -5.22e-05 W/m²

At pressure 1.61 MPa
Experimental burning rate 13.58 mm/s
Calculated burning rate 14.12 mm/s
First Conditional Fuel
Surface temperature 630.9 K
Linear burning rate 18.84 mm/s
Mass decomposition rate 31.25 kg·s⁻¹·m⁻²

Kinetic flame heat flux density to surface 2.48e+06 W/m²
Kinetic flame height 123.91 μm
Average kinetic flame density 2.95 kg/m³
Average kinetic flame temperature 2,168.95 K
Binder sublimation heat flux density 2.48e+06 W/m²
Surface heat balance error -5.78e-04 W/m²

Secondary Conditional Fuel

Surface temperature 602.12 K
Linear burning rate 4.67 mm/s
Mass decomposition rate 7.74 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 41,879.51 W/m²
Kinetic flame height 1,050.34 μm
Average kinetic flame density 7.78 kg/m³
Average kinetic flame temperature 822.06 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 250,121.73 W/m²
Kinetic flame height 255.43 μm
Average kinetic flame density 6.94 kg/m³
Average kinetic flame temperature 921.56 K
Heat flux density due to metal combustion, 71,851.54 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.0038 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 8.35e-05 m
Pore diameter 6.25e-06 m

Diffusion flame heat flux density 142,177.32 W/m²
Diffusion flame height 2,094.48 μm
Heat flux density to surface within skeletal structure 111,214.39 W/m²
Heat flux density to surface outside skeletal structure 27,413.7 W/m²
Total heat flux density to surface 280,805.42 W/m²
Binder sublimation heat flux density 280,805.42 W/m²
Surface heat balance error -1.02e-04 W/m²

<u>*At pressure 2.22 MPa*</u>

Experimental burning rate 17.01 mm/s
Calculated burning rate 16.51 mm/s

First Conditional Fuel

Surface temperature 638.25 K
Linear burning rate 26.36 mm/s
Mass decomposition rate 43.74 kg·s⁻¹·m⁻²
Kinetic flame heat flux density to surface 3.96e+06 W/m²
Kinetic flame height 77.57 μm
Average kinetic flame density 4.06 kg/m³
Average kinetic flame temperature 2,172.62 K
Binder sublimation heat flux density 3.96e+06 W/m²
Surface heat balance error 0.0014 W/m²

Secondary Conditional Fuel

Surface temperature 604.59 K
Linear burning rate 5.29 mm/s
Mass decomposition rate 8.77 kg·s⁻¹·m⁻²

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 55,618.24 W/m²
Kinetic flame height 786.45 μm
Average kinetic flame density 10.71 kg/m³
Average kinetic flame temperature 823.3 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 477,668.18 W/m²

Kinetic flame height 133.23 μm
Average kinetic flame density 9.56 kg/m³
Average kinetic flame temperature 922.8 K
Heat flux density due to metal combustion, 85,601.11 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.0035 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 7e-05 m
Pore diameter 5.81e-06 m
Diffusion flame heat flux density 125,374.09 W/m²
Diffusion flame height 2,373.22 μm
Heat flux density to surface within skeletal structure 188,269.84 W/m²
Heat flux density to surface outside skeletal structure 37,028.13 W/m²
Total heat flux density to surface 350,672.07 W/m²
Binder sublimation heat flux density 350,672.07 W/m²
Surface heat balance error 3.85e-05 W/m²

At pressure 2.83 MPa
Experimental burning rate 20.17 mm/s
Calculated burning rate 19.05 mm/s
First Conditional Fuel
Surface temperature 643.99 K
Linear burning rate 34.1 mm/s
Mass decomposition rate 56.58 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 5.61e+06 W/m²
Kinetic flame height 54.65 μm
Average kinetic flame density 5.17 kg/m³
Average kinetic flame temperature 2,175.5 K
Binder sublimation heat flux density 5.61e+06 W/m²
Surface heat balance error -0.0025 W/m²
Secondary Conditional Fuel
Surface temperature 607.11 K
Linear burning rate 6 mm/s
Mass decomposition rate 9.96 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 66,794.63 W/m²
Kinetic flame height 651.08 μm
Average kinetic flame density 13.64 kg/m³
Average kinetic flame temperature 824.56 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 754,463.09 W/m²
Kinetic flame height 84.02 μm
Average kinetic flame density 12.17 kg/m³
Average kinetic flame temperature 924.06 K
Heat flux density due to metal combustion, 102,199.8 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.0033 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 5.85e-05 m
Pore diameter 5.39e-06 m
Diffusion flame heat flux density 110,386.42 W/m²
Diffusion flame height 2,693.16 μm
Heat flux density to surface within skeletal structure 280,253.41 W/m²
Heat flux density to surface outside skeletal structure 44,943.06 W/m²
Total heat flux density to surface 435,582.9 W/m²
Binder sublimation heat flux density 435,582.9 W/m²
Surface heat balance error -2.21e-04 W/m²

At pressure 3.44 MPa
Experimental burning rate 23.12 mm/s
Calculated burning rate 21.73 mm/s
First Conditional Fuel
Surface temperature 648.73 K
Linear burning rate 42.03 mm/s
Mass decomposition rate 69.72 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 7.4e+06 W/m^2
Kinetic flame height 41.31 μm
Average kinetic flame density 6.28 kg/m^3
Average kinetic flame temperature 2,177.86 K
Binder sublimation heat flux density 7.4e+06 W/m^2
Surface heat balance error -0.0026 W/m^2
Secondary Conditional Fuel
Surface temperature 609.54 K
Linear burning rate 6.77 mm/s
Mass decomposition rate 11.24 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 75,946.63 W/m^2
Kinetic flame height 569.42 μm
Average kinetic flame density 16.56 kg/m^3
Average kinetic flame temperature 825.77 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.07e+06 W/m^2
Kinetic flame height 59.07 μm
Average kinetic flame density 14.78 kg/m^3
Average kinetic flame temperature 925.27 K
Heat flux density due to metal combustion, 121,078.92 W/m^2
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.0031 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 4.93e-05 m
Pore diameter 5.02e-06 m
Diffusion flame heat flux density 97,729.99 W/m^2
Diffusion flame height 3,039.45 μm
Heat flux density to surface within skeletal structure 383,336.86 W/m^2
Heat flux density to surface outside skeletal structure 51,481.88 W/m^2
Total heat flux density to surface 532,548.73 W/m^2
Binder sublimation heat flux density 532,548.73 W/m^2
Surface heat balance error 1e-04 W/m^2

At pressure 4.06 MPa
Experimental burning rate 25.92 mm/s
Calculated burning rate 24.49 mm/s
First Conditional Fuel
Surface temperature 652.77 K
Linear burning rate 50.11 mm/s
Mass decomposition rate 83.12 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 9.33e+06 W/m^2
Kinetic flame height 32.74 μm
Average kinetic flame density 7.38 kg/m^3
Average kinetic flame temperature 2,179.89 K
Binder sublimation heat flux density 9.33e+06 W/m^2
Surface heat balance error 0.0036 W/m^2
Secondary Conditional Fuel
Surface temperature 611.83 K
Linear burning rate 7.58 mm/s

Mass decomposition rate 12.58 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 83,584.67 W/m^2

Kinetic flame height 514.65 µm

Average kinetic flame density 19.47 kg/m^3

Average kinetic flame temperature 826.92 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.41e+06 W/m^2

Kinetic flame height 44.5 µm

Average kinetic flame density 17.37 kg/m^3

Average kinetic flame temperature 926.42 K

Heat flux density due to metal combustion, 141,852.22 W/m^2

Effective thermal conductivity 1.41 W/(m·K)

Conductive thermal conductivity 1.41 W/(m·K)

Radiative thermal conductivity 0.0029 W/(m·K)

Conductive thermal conductivity balance error 9.240271880983641E-10

Skeleton layer thickness 4.2e-05 m

Pore diameter 4.7e-06 m

Diffusion flame heat flux density 87,221.07 W/m^2

Diffusion flame height 3,403.04 µm

Heat flux density to surface within skeletal structure 495,225.72 W/m^2

Heat flux density to surface outside skeletal structure 56,976.7 W/m^2

Total heat flux density to surface 639,423.49 W/m^2

Binder sublimation heat flux density 639,423.49 W/m^2

Surface heat balance error 2.08e-04 W/m^2

At pressure 4.67 MPa

Experimental burning rate 28.6 mm/s

Calculated burning rate 27.33 mm/s

First Conditional Fuel

Surface temperature 656.3 K

Linear burning rate 58.33 mm/s

Mass decomposition rate 96.76 kg·s^{-1}·m^{-2}

Kinetic flame heat flux density to surface 1.14e+07 W/m^2

Kinetic flame height 26.82 µm

Average kinetic flame density 8.49 kg/m^3

Average kinetic flame temperature 2,181.65 K

Binder sublimation heat flux density 1.14e+07 W/m^2

Surface heat balance error -0.003 W/m^2

Secondary Conditional Fuel

Surface temperature 613.96 K

Linear burning rate 8.42 mm/s

Mass decomposition rate 13.97 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 90,093.19 W/m^2

Kinetic flame height 475.1 µm

Average kinetic flame density 22.37 kg/m^3

Average kinetic flame temperature 827.98 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.78e+06 W/m^2

Kinetic flame height 35.14 µm

Average kinetic flame density 19.97 kg/m^3

Average kinetic flame temperature 927.48 K

Heat flux density due to metal combustion, 164,256.47 W/m^2

Effective thermal conductivity 1.41 W/(m·K)

Conductive thermal conductivity 1.41 W/(m·K)

Radiative thermal conductivity 0.0027 W/(m·K)

Conductive thermal conductivity balance error 9.240271880983641E-10

Skeleton layer thickness 3.63e-05 m

Pore diameter 4.42e-06 m
Diffusion flame heat flux density 78,498.66 W/m²
Diffusion flame height 3,778.45 μm
Heat flux density to surface within skeletal structure 614,492.12 W/m²
Heat flux density to surface outside skeletal structure 61,684.81 W/m²
Total heat flux density to surface 754,675.59 W/m²
Binder sublimation heat flux density 754,675.59 W/m²
Surface heat balance error -1.63e-04 W/m²

At pressure 5.28 MPa
Experimental burning rate 31.17 mm/s
Calculated burning rate 30.22 mm/s
First Conditional Fuel
Surface temperature 659.45 K
Linear burning rate 66.67 mm/s
Mass decomposition rate 110.61 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.35e+07 W/m²
Kinetic flame height 22.54 μm
Average kinetic flame density 9.59 kg/m³
Average kinetic flame temperature 2,183.22 K
Binder sublimation heat flux density 1.35e+07 W/m²
Surface heat balance error 0.003 W/m²
Secondary Conditional Fuel
Surface temperature 615.95 K
Linear burning rate 9.27 mm/s
Mass decomposition rate 15.39 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 95,745.15 W/m²
Kinetic flame height 444.98 μm
Average kinetic flame density 25.27 kg/m³
Average kinetic flame temperature 828.98 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 2.18e+06 W/m²
Kinetic flame height 28.7 μm
Average kinetic flame density 22.56 kg/m³
Average kinetic flame temperature 928.48 K
Heat flux density due to metal combustion, 188,107.31 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.0025 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 3.16e-05 m
Pore diameter 4.17e-06 m
Diffusion flame heat flux density 71,211.67 W/m²
Diffusion flame height 4,162.31 μm
Heat flux density to surface within skeletal structure 740,188.51 W/m²
Heat flux density to surface outside skeletal structure 65,791.61 W/m²
Total heat flux density to surface 877,191.79 W/m²
Binder sublimation heat flux density 877,191.79 W/m²
Surface heat balance error 7.06e-05 W/m²

At pressure 5.89 MPa
Experimental burning rate 33.65 mm/s
Calculated burning rate 33.15 mm/s
First Conditional Fuel
Surface temperature 662.28 K
Linear burning rate 75.13 mm/s
Mass decomposition rate 124.65 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.58e+07 W/m²

Kinetic flame height 19.31 μm
Average kinetic flame density 10.7 kg/m^3
Average kinetic flame temperature 2,184.64 K
Binder sublimation heat flux density 1.58e+07 W/m^2
Surface heat balance error 0.0041 W/m^2

Secondary Conditional Fuel

Surface temperature 617.8 K
Linear burning rate 10.14 mm/s
Mass decomposition rate 16.83 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 100,733.52 W/m^2
Kinetic flame height 421.11 μm
Average kinetic flame density 28.16 kg/m^3
Average kinetic flame temperature 829.9 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 2.59e+06 W/m^2
Kinetic flame height 24.05 μm
Average kinetic flame density 25.15 kg/m^3
Average kinetic flame temperature 929.4 K
Heat flux density due to metal combustion, 213,269.91 W/m^2
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.0024 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 2.79e-05 m
Pore diameter 3.96e-06 m

Diffusion flame heat flux density 65,068.35 W/m^2
Diffusion flame height 4,552.44 μm
Heat flux density to surface within skeletal structure 871,637.53 W/m^2
Heat flux density to surface outside skeletal structure 69,429.74 W/m^2
Total heat flux density to surface 1.01e+06 W/m^2
Binder sublimation heat flux density 1.01e+06 W/m^2
Surface heat balance error -2.26e-04 W/m^2

<u>*At pressure 6.5 MPa*</u>

Experimental burning rate 36.06 mm/s
Calculated burning rate 36.11 mm/s

First Conditional Fuel

Surface temperature 664.86 K
Linear burning rate 83.7 mm/s
Mass decomposition rate 138.86 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 1.81e+07 W/m^2
Kinetic flame height 16.8 μm
Average kinetic flame density 11.8 kg/m^3
Average kinetic flame temperature 2,185.93 K
Binder sublimation heat flux density 1.81e+07 W/m^2
Surface heat balance error -7.73e-05 W/m^2

Secondary Conditional Fuel

Surface temperature 619.53 K
Linear burning rate 11.02 mm/s
Mass decomposition rate 18.29 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 105,196.13 W/m^2
Kinetic flame height 401.6 μm
Average kinetic flame density 31.05 kg/m^3
Average kinetic flame temperature 830.77 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 3.02e+06 W/m^2
Kinetic flame height 20.55 μm

Average kinetic flame density 27.73 kg/m³
Average kinetic flame temperature 930.27 K
Heat flux density due to metal combustion, 239,643.09 W/m²
Effective thermal conductivity 1.41 W/(m·K)
Conductive thermal conductivity 1.41 W/(m·K)
Radiative thermal conductivity 0.0023 W/(m·K)
Conductive thermal conductivity balance error 9.240271880983641E-10
Skeleton layer thickness 2.48e-05 m
Pore diameter 3.77e-06 m
Diffusion flame heat flux density 59,838.43 W/m²
Diffusion flame height 4,947.44 μm
Heat flux density to surface within skeletal structure 1.01e+06 W/m²
Heat flux density to surface outside skeletal structure 72,694.44 W/m²
Total heat flux density to surface 1.14e+06 W/m²
Binder sublimation heat flux density 1.14e+06 W/m²
Surface heat balance error -5.18e-04 W/m²

Propellant Composition "Bas_1"

<u>*At pressure 1 MPa*</u>
Experimental burning rate 14.26 mm/s
Calculated burning rate 15.81 mm/s
First Conditional Fuel
Surface temperature 618.13 K
Linear burning rate 11.93 mm/s
Mass decomposition rate 19.8 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.19e+06 W/m²
Kinetic flame height 258.83 μm
Average kinetic flame density 1.84 kg/m³
Average kinetic flame temperature 2,162.56 K
Binder sublimation heat flux density 1.19e+06 W/m²
Surface heat balance error 1.89e-04 W/m²
Secondary Conditional Fuel
Surface temperature 611.33 K
Linear burning rate 8.4 mm/s
Mass decomposition rate 13.94 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 636,077.44 W/m²
Kinetic flame height 275.7 μm
Average kinetic flame density 2.67 kg/m³
Average kinetic flame temperature 1,488.16 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 117,170.57 W/m²
Kinetic flame height 836.11 μm
Average kinetic flame density 3.6 kg/m³
Average kinetic flame temperature 1,101.16 K
Heat flux density due to metal combustion, 166,154.13 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 6.19e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.03e-05 m
Pore diameter 3.84e-11 m
Diffusion flame heat flux density 144,009.83 W/m²
Diffusion flame height 2,121.15 μm
Heat flux density to surface within skeletal structure 65,870.29 W/m²
Heat flux density to surface outside skeletal structure 488,195.5 W/m²
Total heat flux density to surface 698,075.62 W/m²
Binder sublimation heat flux density 698,075.62 W/m²

Surface heat balance error 3.87e-04 W/m²

At pressure 1.61 MPa

Experimental burning rate 19.92 mm/s

Calculated burning rate 20.74 mm/s

First Conditional Fuel

Surface temperature 627.82 K

Linear burning rate 19.42 mm/s

Mass decomposition rate 32.22 $kg \cdot s^{-1} \cdot m^{-2}$

Kinetic flame heat flux density to surface 2.41e+06 W/m²

Kinetic flame height 127.76 μm

Average kinetic flame density 2.95 kg/m³

Average kinetic flame temperature 2,167.41 K

Binder sublimation heat flux density 2.41e+06 W/m²

Surface heat balance error 9.05e-04 W/m²

Secondary Conditional Fuel

Surface temperature 615.47 K

Linear burning rate 10.42 mm/s

Mass decomposition rate 17.28 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 937,335.8 W/m²

Kinetic flame height 186.65 μm

Average kinetic flame density 4.29 kg/m³

Average kinetic flame temperature 1,490.24 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 319,229.19 W/m²

Kinetic flame height 305.59 μm

Average kinetic flame density 5.8 kg/m³

Average kinetic flame temperature 1,103.24 K

Heat flux density due to metal combustion, 224,572.99 W/m²

Effective thermal conductivity 0.4 W/(m·K)

Conductive thermal conductivity 0.4 W/(m·K)

Radiative thermal conductivity 5.45e-08 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 7.57e-06 m

Pore diameter 3.39e-11 m

Diffusion flame heat flux density 116,024.8 W/m²

Diffusion flame height 2,629.2 μm

Heat flux density to surface within skeletal structure 111,547.21 W/m²

Heat flux density to surface outside skeletal structure 745,065.16 W/m²

Total heat flux density to surface 972,637.16 W/m²

Binder sublimation heat flux density 972,637.16 W/m²

Surface heat balance error 4.48e-04 W/m²

At pressure 2.22 MPa

Experimental burning rate 24.95 mm/s

Calculated burning rate 25.13 mm/s

First Conditional Fuel

Surface temperature 634.69 K

Linear burning rate 27.19 mm/s

Mass decomposition rate 45.11 $kg \cdot s^{-1} \cdot m^{-2}$

Kinetic flame heat flux density to surface 3.84e+06 W/m²

Kinetic flame height 80.02 μm

Average kinetic flame density 4.06 kg/m³

Average kinetic flame temperature 2,170.85 K

Binder sublimation heat flux density 3.84e+06 W/m²

Surface heat balance error 0.0012 W/m²

Secondary Conditional Fuel

Surface temperature 618.63 K

Linear burning rate 12.24 mm/s
Mass decomposition rate 20.31 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.2e+06 W/m^2
Kinetic flame height 145.88 μm
Average kinetic flame density 5.91 kg/m^3
Average kinetic flame temperature 1,491.81 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 618,751.64 W/m^2
Kinetic flame height 157.15 μm
Average kinetic flame density 7.98 kg/m^3
Average kinetic flame temperature 1,104.81 K
Heat flux density due to metal combustion, 281,757.18 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.96e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 6.02e-06 m
Pore diameter 3.08e-11 m
Diffusion flame heat flux density 98,601.02 W/m^2
Diffusion flame height 3,090.61 μm
Heat flux density to surface within skeletal structure 170,757.19 W/m^2
Heat flux density to surface outside skeletal structure 970,136.03 W/m^2
Total heat flux density to surface 1.24e+06 W/m^2
Binder sublimation heat flux density 1.24e+06 W/m^2
Surface heat balance error -4.68e-04 W/m^2

At pressure 2.83 MPa
Experimental burning rate 29.57 mm/s
Calculated burning rate 29.2 mm/s
First Conditional Fuel
Surface temperature 640.06 K
Linear burning rate 35.19 mm/s
Mass decomposition rate 58.37 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 5.44e+06 W/m^2
Kinetic flame height 56.39 μm
Average kinetic flame density 5.17 kg/m^3
Average kinetic flame temperature 2,173.53 K
Binder sublimation heat flux density 5.44e+06 W/m^2
Surface heat balance error 0.0015 W/m^2
Secondary Conditional Fuel
Surface temperature 621.2 K
Linear burning rate 13.95 mm/s
Mass decomposition rate 23.14 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.43e+06 W/m^2
Kinetic flame height 122.07 μm
Average kinetic flame density 7.53 kg/m^3
Average kinetic flame temperature 1,493.1 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.01e+06 W/m^2
Kinetic flame height 95.71 μm
Average kinetic flame density 10.17 kg/m^3
Average kinetic flame temperature 1,106.1 K
Heat flux density due to metal combustion, 338,286.84 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.59e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 5.01e-06 m
Pore diameter 2.85e-11 m
Diffusion flame heat flux density 86,478.73 W/m²
Diffusion flame height 3,520.87 μm
Heat flux density to surface within skeletal structure 242,254.48 W/m²
Heat flux density to surface outside skeletal structure 1.17e+06 W/m²
Total heat flux density to surface 1.5e+06 W/m²
Binder sublimation heat flux density 1.5e+06 W/m²
Surface heat balance error -2.81e-04 W/m²

At pressure 3.44 MPa
Experimental burning rate 33.9 mm/s
Calculated burning rate 33.04 mm/s
First Conditional Fuel
Surface temperature 644.48 K
Linear burning rate 43.37 mm/s
Mass decomposition rate 71.95 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 7.18e+06 W/m²
Kinetic flame height 42.65 μm
Average kinetic flame density 6.28 kg/m³
Average kinetic flame temperature 2,175.74 K
Binder sublimation heat flux density 7.18e+06 W/m²
Surface heat balance error -0.0018 W/m²
Secondary Conditional Fuel
Surface temperature 623.38 K
Linear burning rate 15.57 mm/s
Mass decomposition rate 25.82 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.64e+06 W/m²
Kinetic flame height 106.3 μm
Average kinetic flame density 9.15 kg/m³
Average kinetic flame temperature 1,494.19 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.5e+06 W/m²
Kinetic flame height 64.5 μm
Average kinetic flame density 12.35 kg/m³
Average kinetic flame temperature 1,107.19 K
Heat flux density due to metal combustion, 394,608.49 W/m²
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 4.3e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 4.29e-06 m
Pore diameter 2.67e-11 m
Diffusion flame heat flux density 77,431.85 W/m²
Diffusion flame height 3,929.42 μm
Heat flux density to surface within skeletal structure 325,224.99 W/m²
Heat flux density to surface outside skeletal structure 1.36e+06 W/m²
Total heat flux density to surface 1.76e+06 W/m²
Binder sublimation heat flux density 1.76e+06 W/m²
Surface heat balance error -4.15e-04 W/m²

At pressure 4.06 MPa
Experimental burning rate 38.01 mm/s
Calculated burning rate 36.72 mm/s
First Conditional Fuel
Surface temperature 648.25 K
Linear burning rate 51.72 mm/s
Mass decomposition rate 85.8 $kg \cdot s^{-1} \cdot m^{-2}$

- Kinetic flame heat flux density to surface 9.05e+06 W/m^2
- Kinetic flame height 33.8 μm
- Average kinetic flame density 7.39 kg/m^3
- Average kinetic flame temperature 2,177.63 K
- Binder sublimation heat flux density 9.05e+06 W/m^2
- Surface heat balance error -0.0013 W/m^2

Secondary Conditional Fuel

- Surface temperature 625.28 K
- Linear burning rate 17.12 mm/s
- Mass decomposition rate 28.41 $kg\cdot s^{-1}\cdot m^{-2}$
- *Region Outside Skeleton Structure*
  - Kinetic flame heat flux density to surface 1.83e+06 W/m^2
  - Kinetic flame height 95.01 μm
  - Average kinetic flame density 10.77 kg/m^3
  - Average kinetic flame temperature 1,495.14 K
- *Skeleton Structure Region*
  - Kinetic flame heat flux density to surface 2.08e+06 W/m^2
  - Kinetic flame height 46.5 μm
  - Average kinetic flame density 14.53 kg/m^3
  - Average kinetic flame temperature 1,108.14 K
  - Heat flux density due to metal combustion, 451,036.84 W/m^2
  - Effective thermal conductivity 0.4 $W/(m\cdot K)$
  - Conductive thermal conductivity 0.4 $W/(m\cdot K)$
  - Radiative thermal conductivity 4.07e-08 $W/(m\cdot K)$
  - Conductive thermal conductivity balance error -4.092294297874943E-08
  - Skeleton layer thickness 3.75e-06 m
  - Pore diameter 2.53e-11 m
- Diffusion flame heat flux density 70,350.19 W/m^2
- Diffusion flame height 4,322.26 μm
- Heat flux density to surface within skeletal structure 419,021.39 W/m^2
- Heat flux density to surface outside skeletal structure 1.53e+06 W/m^2
- Total heat flux density to surface 2.02e+06 W/m^2
- Binder sublimation heat flux density 2.02e+06 W/m^2
- Surface heat balance error 7.18e-04 W/m^2

At pressure 4.67 MPa

Experimental burning rate 41.93 mm/s
Calculated burning rate 40.29 mm/s

First Conditional Fuel

- Surface temperature 651.55 K
- Linear burning rate 60.22 mm/s
- Mass decomposition rate 99.9 $kg\cdot s^{-1}\cdot m^{-2}$
- Kinetic flame heat flux density to surface 1.1e+07 W/m^2
- Kinetic flame height 27.7 μm
- Average kinetic flame density 8.5 kg/m^3
- Average kinetic flame temperature 2,179.27 K
- Binder sublimation heat flux density 1.1e+07 W/m^2
- Surface heat balance error -0.0023 W/m^2

Secondary Conditional Fuel

- Surface temperature 626.98 K
- Linear burning rate 18.63 mm/s
- Mass decomposition rate 30.91 $kg\cdot s^{-1}\cdot m^{-2}$
- *Region Outside Skeleton Structure*
  - Kinetic flame heat flux density to surface 2.01e+06 W/m^2
  - Kinetic flame height 86.5 μm
  - Average kinetic flame density 12.38 kg/m^3
  - Average kinetic flame temperature 1,495.99 K
- *Skeleton Structure Region*
  - Kinetic flame heat flux density to surface 2.74e+06 W/m^2

Kinetic flame height 35.18 μm
Average kinetic flame density 16.7 kg/m^3
Average kinetic flame temperature 1,108.99 K
Heat flux density due to metal combustion, 507,796.86 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 3.87e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.32e-06 m
Pore diameter 2.4e-11 m
Diffusion flame heat flux density 64,613.04 W/m^2
Diffusion flame height 4,703.42 μm
Heat flux density to surface within skeletal structure 523,095.39 W/m^2
Heat flux density to surface outside skeletal structure 1.69e+06 W/m^2
Total heat flux density to surface 2.27e+06 W/m^2
Binder sublimation heat flux density 2.27e+06 W/m^2
Surface heat balance error -5.99e-04 W/m^2

At pressure 5.28 MPa
Experimental burning rate 45.71 mm/s
Calculated burning rate 43.76 mm/s
First Conditional Fuel
Surface temperature 654.48 K
Linear burning rate 68.85 mm/s
Mass decomposition rate 114.22 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.31e+07 W/m^2
Kinetic flame height 23.28 μm
Average kinetic flame density 9.61 kg/m^3
Average kinetic flame temperature 2,180.74 K
Binder sublimation heat flux density 1.31e+07 W/m^2
Surface heat balance error -0.0041 W/m^2
Secondary Conditional Fuel
Surface temperature 628.52 K
Linear burning rate 20.11 mm/s
Mass decomposition rate 33.36 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.18e+06 W/m^2
Kinetic flame height 79.82 μm
Average kinetic flame density 14 kg/m^3
Average kinetic flame temperature 1,496.76 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 3.49e+06 W/m^2
Kinetic flame height 27.58 μm
Average kinetic flame density 18.88 kg/m^3
Average kinetic flame temperature 1,109.76 K
Heat flux density due to metal combustion, 565,046.89 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 3.7e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.98e-06 m
Pore diameter 2.3e-11 m
Diffusion flame heat flux density 59,843.97 W/m^2
Diffusion flame height 5,075.67 μm
Heat flux density to surface within skeletal structure 636,861.83 W/m^2
Heat flux density to surface outside skeletal structure 1.83e+06 W/m^2
Total heat flux density to surface 2.53e+06 W/m^2
Binder sublimation heat flux density 2.53e+06 W/m^2
Surface heat balance error -6.54e-04 W/m^2

At pressure 5.89 MPa
Experimental burning rate 49.35 mm/s
Calculated burning rate 47.17 mm/s
First Conditional Fuel
Surface temperature 657.12 K
Linear burning rate 77.6 mm/s
Mass decomposition rate 128.74 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.53e+07 W/m^2
Kinetic flame height 19.95 µm
Average kinetic flame density 10.71 kg/m^3
Average kinetic flame temperature 2,182.06 K
Binder sublimation heat flux density 1.53e+07 W/m^2
Surface heat balance error 0.0017 W/m^2
Secondary Conditional Fuel
Surface temperature 629.93 K
Linear burning rate 21.55 mm/s
Mass decomposition rate 35.76 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.33e+06 W/m^2
Kinetic flame height 74.44 µm
Average kinetic flame density 15.61 kg/m^3
Average kinetic flame temperature 1,497.46 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 4.32e+06 W/m^2
Kinetic flame height 22.25 µm
Average kinetic flame density 21.05 kg/m^3
Average kinetic flame temperature 1,110.46 K
Heat flux density due to metal combustion, 622,931.2 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 3.55e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.71e-06 m
Pore diameter 2.21e-11 m
Diffusion flame heat flux density 55,797.8 W/m^2
Diffusion flame height 5,441.2 µm
Heat flux density to surface within skeletal structure 760,195.81 W/m^2
Heat flux density to surface outside skeletal structure 1.97e+06 W/m^2
Total heat flux density to surface 2.79e+06 W/m^2
Binder sublimation heat flux density 2.79e+06 W/m^2
Surface heat balance error -9.88e-04 W/m^2

At pressure 6.5 MPa
Experimental burning rate 52.88 mm/s
Calculated burning rate 50.53 mm/s
First Conditional Fuel
Surface temperature 659.52 K
Linear burning rate 86.46 mm/s
Mass decomposition rate 143.44 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.76e+07 W/m^2
Kinetic flame height 17.36 µm
Average kinetic flame density 11.82 kg/m^3
Average kinetic flame temperature 2,183.26 K
Binder sublimation heat flux density 1.76e+07 W/m^2
Surface heat balance error -0.0078 W/m^2
Secondary Conditional Fuel
Surface temperature 631.24 K
Linear burning rate 22.98 mm/s

Mass decomposition rate 38.13 kg·s^{-1}·m^{-2}
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.48e+06 W/m^2
Kinetic flame height 70 μm
Average kinetic flame density 17.22 kg/m^3
Average kinetic flame temperature 1,498.12 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 5.23e+06 W/m^2
Kinetic flame height 18.35 μm
Average kinetic flame density 23.22 kg/m^3
Average kinetic flame temperature 1,111.12 K
Heat flux density due to metal combustion, 681,524.9 W/m^2
Effective thermal conductivity 0.4 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 3.42e-08 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.47e-06 m
Pore diameter 2.12e-11 m
Diffusion flame heat flux density 52,310.69 W/m^2
Diffusion flame height 5,801.42 μm
Heat flux density to surface within skeletal structure 892,397.56 W/m^2
Heat flux density to surface outside skeletal structure 2.1e+06 W/m^2
Total heat flux density to surface 3.05e+06 W/m^2
Binder sublimation heat flux density 3.05e+06 W/m^2
Surface heat balance error -0.0014 W/m^2

Propellant Composition "Bas_0"

<u>*At pressure 1 MPa*</u>
Experimental burning rate 5.97 mm/s
Calculated burning rate 6.05 mm/s
First Conditional Fuel
Surface temperature 838.46 K
Linear burning rate 4.54 mm/s
Mass decomposition rate 7.53 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 2.94e+06 W/m^2
Kinetic flame height 97.44 μm
Average kinetic flame density 1.75 kg/m^3
Average kinetic flame temperature 2,272.73 K
Binder sublimation heat flux density 2.94e+06 W/m^2
Surface heat balance error 2.09e-04 W/m^2
Secondary Conditional Fuel
Surface temperature 829.68 K
Linear burning rate 3.22 mm/s
Mass decomposition rate 5.34 kg·s^{-1}·m^{-2}
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 1.75e+06 W/m^2
Kinetic flame height 87.61 μm
Average kinetic flame density 2.48 kg/m^3
Average kinetic flame temperature 1,597.34 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.59e+06 W/m^2
Kinetic flame height 48.03 μm
Average kinetic flame density 3.28 kg/m^3
Average kinetic flame temperature 1,210.34 K
Heat flux density due to metal combustion, 22,189.93 W/m^2
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.017 W/(m·K)

Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 6.97e-05 m
Pore diameter 1.08e-05 m
Diffusion flame heat flux density 349,046.32 W/m²
Diffusion flame height 812.59 µm
Heat flux density to surface within skeletal structure 942,050.99 W/m²
Heat flux density to surface outside skeletal structure 725,415.69 W/m²
Total heat flux density to surface 2.02e+06 W/m²
Binder sublimation heat flux density 2.02e+06 W/m²
Surface heat balance error 9.63e-05 W/m²

At pressure 1.61 MPa
Experimental burning rate 7.9 mm/s
Calculated burning rate 7.84 mm/s
First Conditional Fuel
Surface temperature 853.31 K
Linear burning rate 8 mm/s
Mass decomposition rate 13.27 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 5.48e+06 W/m²
Kinetic flame height 52.07 µm
Average kinetic flame density 2.8 kg/m³
Average kinetic flame temperature 2,280.16 K
Binder sublimation heat flux density 5.48e+06 W/m²
Surface heat balance error 3.25e-04 W/m²
Secondary Conditional Fuel
Surface temperature 834.32 K
Linear burning rate 3.86 mm/s
Mass decomposition rate 6.41 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 2.66e+06 W/m²
Kinetic flame height 57.46 µm
Average kinetic flame density 4 kg/m³
Average kinetic flame temperature 1,599.66 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.68e+06 W/m²
Kinetic flame height 44.92 µm
Average kinetic flame density 5.27 kg/m³
Average kinetic flame temperature 1,212.66 K
Heat flux density due to metal combustion, 28,527.48 W/m²
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.016 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 5.39e-05 m
Pore diameter 9.73e-06 m
Diffusion flame heat flux density 290,246.98 W/m²
Diffusion flame height 975.61 µm
Heat flux density to surface within skeletal structure 879,708.24 W/m²
Heat flux density to surface outside skeletal structure 1.3e+06 W/m²
Total heat flux density to surface 2.47e+06 W/m²
Binder sublimation heat flux density 2.47e+06 W/m²
Surface heat balance error -4.81e-04 W/m²

At pressure 2.22 MPa
Experimental burning rate 9.56 mm/s
Calculated burning rate 9.43 mm/s
First Conditional Fuel
Surface temperature 863.63 K
Linear burning rate 11.72 mm/s

Mass decomposition rate 19.44 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 8.33e+06 W/m^2
Kinetic flame height 34.13 μm
Average kinetic flame density 3.86 kg/m^3
Average kinetic flame temperature 2,285.32 K
Binder sublimation heat flux density 8.33e+06 W/m^2
Surface heat balance error 4.14e-04 W/m^2

Secondary Conditional Fuel

Surface temperature 838.11 K
Linear burning rate 4.48 mm/s
Mass decomposition rate 7.43 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 3.45e+06 W/m^2
Kinetic flame height 44.29 μm
Average kinetic flame density 5.51 kg/m^3
Average kinetic flame temperature 1,601.56 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.71e+06 W/m^2
Kinetic flame height 43.96 μm
Average kinetic flame density 7.26 kg/m^3
Average kinetic flame temperature 1,214.56 K
Heat flux density due to metal combustion, 34,964.44 W/m^2
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.014 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 4.37e-05 m
Pore diameter 8.92e-06 m

Diffusion flame heat flux density 250,014.59 W/m^2
Diffusion flame height 1,131.09 μm
Heat flux density to surface within skeletal structure 818,760.36 W/m^2
Heat flux density to surface outside skeletal structure 1.83e+06 W/m^2
Total heat flux density to surface 2.9e+06 W/m^2
Binder sublimation heat flux density 2.9e+06 W/m^2
Surface heat balance error -2.38e-04 W/m^2

<u>*At pressure 2.83 MPa*</u>

Experimental burning rate 11.03 mm/s
Calculated burning rate 10.9 mm/s

First Conditional Fuel

Surface temperature 871.6 K
Linear burning rate 15.64 mm/s
Mass decomposition rate 25.94 kg·s^{-1}·m^{-2}
Kinetic flame heat flux density to surface 1.14e+07 W/m^2
Kinetic flame height 24.81 μm
Average kinetic flame density 4.91 kg/m^3
Average kinetic flame temperature 2,289.3 K
Binder sublimation heat flux density 1.14e+07 W/m^2
Surface heat balance error 0.002 W/m^2

Secondary Conditional Fuel

Surface temperature 841.29 K
Linear burning rate 5.07 mm/s
Mass decomposition rate 8.41 kg·s^{-1}·m^{-2}

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 4.14e+06 W/m^2
Kinetic flame height 36.79 μm
Average kinetic flame density 7.01 kg/m^3
Average kinetic flame temperature 1,603.15 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.71e+06 W/m²
Kinetic flame height 43.72 μm
Average kinetic flame density 9.25 kg/m³
Average kinetic flame temperature 1,216.15 K
Heat flux density due to metal combustion, 41,417.79 W/m²
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.013 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.67e-05 m
Pore diameter 8.3e-06 m

Diffusion flame heat flux density 220,835.18 W/m²
Diffusion flame height 1,279.1 μm
Heat flux density to surface within skeletal structure 767,220.67 W/m²
Heat flux density to surface outside skeletal structure 2.33e+06 W/m²
Total heat flux density to surface 3.32e+06 W/m²
Binder sublimation heat flux density 3.32e+06 W/m²
Surface heat balance error -1.68e-04 W/m²

At pressure 3.44 MPa

Experimental burning rate 12.38 mm/s
Calculated burning rate 12.28 mm/s

First Conditional Fuel

Surface temperature 878.12 K
Linear burning rate 19.72 mm/s
Mass decomposition rate 32.72 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.47e+07 W/m²
Kinetic flame height 19.2 μm
Average kinetic flame density 5.96 kg/m³
Average kinetic flame temperature 2,292.56 K
Binder sublimation heat flux density 1.47e+07 W/m²
Surface heat balance error 0.0031 W/m²

Secondary Conditional Fuel

Surface temperature 844.02 K
Linear burning rate 5.63 mm/s
Mass decomposition rate 9.33 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 4.77e+06 W/m²
Kinetic flame height 31.87 μm
Average kinetic flame density 8.52 kg/m³
Average kinetic flame temperature 1,604.51 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.71e+06 W/m²
Kinetic flame height 43.76 μm
Average kinetic flame density 11.23 kg/m³
Average kinetic flame temperature 1,217.51 K
Heat flux density due to metal combustion, 47,848.68 W/m²
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.013 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 3.17e-05 m
Pore diameter 7.8e-06 m

Diffusion flame heat flux density 198,688.69 W/m²
Diffusion flame height 1,420.3 μm
Heat flux density to surface within skeletal structure 724,417.08 W/m²
Heat flux density to surface outside skeletal structure 2.8e+06 W/m²
Total heat flux density to surface 3.73e+06 W/m²
Binder sublimation heat flux density 3.73e+06 W/m²

Surface heat balance error -8.34e-04 W/m²

At pressure 4.06 MPa
Experimental burning rate 13.63 mm/s
Calculated burning rate 13.59 mm/s
First Conditional Fuel
Surface temperature 883.64 K
Linear burning rate 23.95 mm/s
Mass decomposition rate 39.73 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 1.82e+07 W/m²
Kinetic flame height 15.5 µm
Average kinetic flame density 7.01 kg/m³
Average kinetic flame temperature 2,295.32 K
Binder sublimation heat flux density 1.82e+07 W/m²
Surface heat balance error 0.0053 W/m²
Secondary Conditional Fuel
Surface temperature 846.4 K
Linear burning rate 6.16 mm/s
Mass decomposition rate 10.22 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 5.35e+06 W/m²
Kinetic flame height 28.36 µm
Average kinetic flame density 10.02 kg/m³
Average kinetic flame temperature 1,605.7 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.69e+06 W/m²
Kinetic flame height 43.94 µm
Average kinetic flame density 13.21 kg/m³
Average kinetic flame temperature 1,218.7 K
Heat flux density due to metal combustion, 54,239.68 W/m²
Effective thermal conductivity 0.42 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.012 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.78e-05 m
Pore diameter 7.39e-06 m
Diffusion flame heat flux density 181,271.88 W/m²
Diffusion flame height 1,555.45 µm
Heat flux density to surface within skeletal structure 688,647.19 W/m²
Heat flux density to surface outside skeletal structure 3.25e+06 W/m²
Total heat flux density to surface 4.12e+06 W/m²
Binder sublimation heat flux density 4.12e+06 W/m²
Surface heat balance error -4.95e-04 W/m²

At pressure 4.67 MPa
Experimental burning rate 14.8 mm/s
Calculated burning rate 14.85 mm/s
First Conditional Fuel
Surface temperature 888.45 K
Linear burning rate 28.3 mm/s
Mass decomposition rate 46.95 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.19e+07 W/m²
Kinetic flame height 12.89 µm
Average kinetic flame density 8.06 kg/m³
Average kinetic flame temperature 2,297.72 K
Binder sublimation heat flux density 2.19e+07 W/m²
Surface heat balance error 0.0034 W/m²
Secondary Conditional Fuel
Surface temperature 848.51 K

Linear burning rate 6.68 mm/s
Mass decomposition rate 11.08 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 5.9e+06 W/m^2
Kinetic flame height 25.7 μm
Average kinetic flame density 11.53 kg/m^3
Average kinetic flame temperature 1,606.76 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.68e+06 W/m^2
Kinetic flame height 44.17 μm
Average kinetic flame density 15.18 kg/m^3
Average kinetic flame temperature 1,219.76 K
Heat flux density due to metal combustion, 60,583.4 W/m^2
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.011 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.49e-05 m
Pore diameter 7.05e-06 m
Diffusion flame heat flux density 167,184.01 W/m^2
Diffusion flame height 1,685.26 μm
Heat flux density to surface within skeletal structure 658,398.49 W/m^2
Heat flux density to surface outside skeletal structure 3.67e+06 W/m^2
Total heat flux density to surface 4.5e+06 W/m^2
Binder sublimation heat flux density 4.5e+06 W/m^2
Surface heat balance error -4.47e-04 W/m^2

<u>*At pressure 5.28 MPa*</u>
Experimental burning rate 15.92 mm/s
Calculated burning rate 16.06 mm/s
First Conditional Fuel
Surface temperature 892.7 K
Linear burning rate 32.76 mm/s
Mass decomposition rate 54.35 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.57e+07 W/m^2
Kinetic flame height 10.97 μm
Average kinetic flame density 9.11 kg/m^3
Average kinetic flame temperature 2,299.85 K
Binder sublimation heat flux density 2.57e+07 W/m^2
Surface heat balance error -0.006 W/m^2
Secondary Conditional Fuel
Surface temperature 850.41 K
Linear burning rate 7.17 mm/s
Mass decomposition rate 11.9 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 6.41e+06 W/m^2
Kinetic flame height 23.61 μm
Average kinetic flame density 13.03 kg/m^3
Average kinetic flame temperature 1,607.7 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.67e+06 W/m^2
Kinetic flame height 44.44 μm
Average kinetic flame density 17.16 kg/m^3
Average kinetic flame temperature 1,220.7 K
Heat flux density due to metal combustion, 66,877.19 W/m^2
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.011 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08

Skeleton layer thickness 2.25e-05 m
Pore diameter 6.76e-06 m
Diffusion flame heat flux density 155,528.58 W/m²
Diffusion flame height 1,810.34 μm
Heat flux density to surface within skeletal structure 632,505.08 W/m²
Heat flux density to surface outside skeletal structure 4.07e+06 W/m²
Total heat flux density to surface 4.86e+06 W/m²
Binder sublimation heat flux density 4.86e+06 W/m²
Surface heat balance error 0.0017 W/m²

At pressure 5.89 MPa
Experimental burning rate 16.98 mm/s
Calculated burning rate 17.22 mm/s
First Conditional Fuel
Surface temperature 896.53 K
Linear burning rate 37.32 mm/s
Mass decomposition rate 61.91 $kg \cdot s^{-1} \cdot m^{-2}$
Kinetic flame heat flux density to surface 2.96e+07 W/m²
Kinetic flame height 9.5 μm
Average kinetic flame density 10.15 kg/m³
Average kinetic flame temperature 2,301.76 K
Binder sublimation heat flux density 2.96e+07 W/m²
Surface heat balance error 0.0061 W/m²
Secondary Conditional Fuel
Surface temperature 852.12 K
Linear burning rate 7.65 mm/s
Mass decomposition rate 12.69 $kg \cdot s^{-1} \cdot m^{-2}$
Region Outside Skeleton Structure
Kinetic flame heat flux density to surface 6.91e+06 W/m²
Kinetic flame height 21.91 μm
Average kinetic flame density 14.53 kg/m³
Average kinetic flame temperature 1,608.56 K
Skeleton Structure Region
Kinetic flame heat flux density to surface 1.65e+06 W/m²
Kinetic flame height 44.71 μm
Average kinetic flame density 19.13 kg/m³
Average kinetic flame temperature 1,221.56 K
Heat flux density due to metal combustion, 73,120.76 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.01 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 2.05e-05 m
Pore diameter 6.51e-06 m
Diffusion flame heat flux density 145,706.05 W/m²
Diffusion flame height 1,931.2 μm
Heat flux density to surface within skeletal structure 610,086.98 W/m²
Heat flux density to surface outside skeletal structure 4.46e+06 W/m²
Total heat flux density to surface 5.22e+06 W/m²
Binder sublimation heat flux density 5.22e+06 W/m²
Surface heat balance error 3.03e-04 W/m²

At pressure 6.5 MPa
Experimental burning rate 18 mm/s
Calculated burning rate 18.35 mm/s
First Conditional Fuel
Surface temperature 900 K
Linear burning rate 41.97 mm/s
Mass decomposition rate 69.62 $kg \cdot s^{-1} \cdot m^{-2}$

Kinetic flame heat flux density to surface 3.36e+07 W/m²
Kinetic flame height 8.35 μm
Average kinetic flame density 11.2 kg/m³
Average kinetic flame temperature 2,303.5 K
Binder sublimation heat flux density 3.36e+07 W/m²
Surface heat balance error -0.0054 W/m²

Secondary Conditional Fuel

Surface temperature 853.69 K
Linear burning rate 8.11 mm/s
Mass decomposition rate 13.46 $kg \cdot s^{-1} \cdot m^{-2}$

Region Outside Skeleton Structure

Kinetic flame heat flux density to surface 7.37e+06 W/m²
Kinetic flame height 20.49 μm
Average kinetic flame density 16.03 kg/m³
Average kinetic flame temperature 1,609.35 K

Skeleton Structure Region

Kinetic flame heat flux density to surface 1.64e+06 W/m²
Kinetic flame height 44.98 μm
Average kinetic flame density 21.11 kg/m³
Average kinetic flame temperature 1,222.35 K
Heat flux density due to metal combustion, 79,314.93 W/m²
Effective thermal conductivity 0.41 W/(m·K)
Conductive thermal conductivity 0.4 W/(m·K)
Radiative thermal conductivity 0.01 W/(m·K)
Conductive thermal conductivity balance error -4.092294297874943E-08
Skeleton layer thickness 1.89e-05 m
Pore diameter 6.29e-06 m

Diffusion flame heat flux density 137,300.66 W/m²
Diffusion flame height 2,048.28 μm
Heat flux density to surface within skeletal structure 590,483.99 W/m²
Heat flux density to surface outside skeletal structure 4.84e+06 W/m²
Total heat flux density to surface 5.57e+06 W/m²
Binder sublimation heat flux density 5.57e+06 W/m²
Surface heat balance error 7.34e-04 W/m²

1 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	16.94 mm/s		15.24 mm/s		9.73 mm/s		14.26 mm/s		5.97 mm/s	
Calculated Burning Rate	15.15 mm/s		15.47 mm/s		11.79 mm/s		15.81 mm/s		6.05 mm/s	
Difference	1.79 mm/s		-0.22 mm/s		-2.06 mm/s		-1.54 mm/s		-0.08 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	11.58 mm/s	8.03 mm/s	11.58 mm/s	11.58 mm/s	11.58 mm/s	4.19 mm/s	11.93 mm/s	8.4 mm/s	4.54 mm/s	3.22 mm/s
Surface Temperature	620.57 K	612.98 K	620.57 K	620.57 K	620.57 K	600 K	618.13 K	611.33 K	838.46 K	829.68 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.23e+06 W/m^2		1.23e+06 W/m^2		1.23e+06 W/m^2		1.19e+06 W/m^2		2.94e+06 W/m^2	
Kinetic Flame Height (Inter-Pocket)	251.2 μm		251.2 μm		251.2 μm		258.83 μm		97.44 μm	
Kinetic Flame Heat Flux Density (Skeleton)	134,764.9 W/m^2		435,508.41 W/m^2		88,801.83 W/m^2		117,170.57 W/m^2		1.59e+06 W/m^2	
Kinetic Flame Height (Skeleton)	725.72 μm		459.79 μm		721.83 μm		836.11 μm		48.03 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	666,404.95 W/m^2		462,901.24 W/m^2		25,510.21 W/m^2		636,077.44 W/m^2		1.75e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	262.91 μm		376.85 μm		1,732.64 μm		275.7 μm		87.61 μm	
Diffusion Flame Heat Flux Density	147,643.25 W/m^2		761,170.48 W/m^2		158,543.34 W/m^2		144,009.83 W/m^2		349,046.32 W/m^2	
Diffusion Flame Height	2,025.84 μm		401.15 μm		1,879.61 μm		2,121.15 μm		812.59 μm	
Metal Combustion Heat Flux Density	44,657.46 W/m^2		46,915.98 W/m^2		61,738.89 W/m^2		166,154.13 W/m^2		22,189.93 W/m^2	

1.61 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	21.71 mm/s		20.68 mm/s		13.58 mm/s		19.92 mm/s		7.9 mm/s	
Calculated Burning Rate	19.89 mm/s		20.32 mm/s		14.12 mm/s		20.74 mm/s		7.84 mm/s	
Difference	1.82 mm/s		0.36 mm/s		-0.54 mm/s		-0.82 mm/s		0.064 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	18.84 mm/s	9.96 mm/s	18.84 mm/s	13.29 mm/s	18.84 mm/s	4.67 mm/s	19.42 mm/s	10.42 mm/s	8 mm/s	3.86 mm/s
Surface Temperature	630.9 K	617.43 K	630.9 K	623.45 K	630.9 K	602.12 K	627.82 K	615.47 K	853.31 K	834.32 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	2.48e+06 W/m²		2.48e+06 W/m²		2.48e+06 W/m²		2.41e+06 W/m²		5.48e+06 W/m²	
Kinetic Flame Height (Inter-Pocket)	123.91 µm		123.91 µm		123.91 µm		127.76 µm		52.07 µm	
Kinetic Flame Heat Flux Density (Skeleton)	340,712.96 W/m²		1.19e+06 W/m²		250,121.73 W/m²		319,229.19 W/m²		1.68e+06 W/m²	
Kinetic Flame Height (Skeleton)	285.74 µm		168.14 µm		255.43 µm		305.59 µm		44.92 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	980,450.85 W/m²		736,509.9 W/m²		41,879.51 W/m²		937,335.8 W/m²		2.66e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	178.24 µm		236.46 µm		1,050.34 µm		186.65 µm		57.46 µm	
Diffusion Flame Heat Flux Density	118,733.63 W/m²		662,723.79 W/m²		142,177.32 W/m²		116,024.8 W/m²		290,246.98 W/m²	
Diffusion Flame Height	2,515.35 µm		460.3 µm		2,094.48 µm		2,629.2 µm		975.61 µm	
Metal Combustion Heat Flux Density	60,364.74 W/m²		56,273.1 W/m²		71,851.54 W/m²		224,572.99 W/m²		28,527.48 W/m²	

2.22 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	25.67 mm/s		25.41 mm/s		17.01 mm/s		24.95 mm/s		9.56 mm/s	
Calculated Burning Rate	24.09 mm/s		24.72 mm/s		16.51 mm/s		25.13 mm/s		9.43 mm/s	
Difference	1.58 mm/s		0.69 mm/s		0.5 mm/s		-0.19 mm/s		0.13 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	26.36 mm/s	11.71 mm/s	26.36 mm/s	15.12 mm/s	26.36 mm/s	5.29 mm/s	27.19 mm/s	12.24 mm/s	11.72 mm/s	4.48 mm/s
Surface Temperature	638.25 K	620.79 K	638.25 K	626.18 K	638.25 K	604.59 K	634.69 K	618.63 K	863.63 K	838.11 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	3.96e+06 W/m^2		3.96e+06 W/m^2		3.96e+06 W/m^2		3.84e+06 W/m^2		8.33e+06 W/m^2	
Kinetic Flame Height (Inter-Pocket)	77.57 µm		77.57 µm		77.57 µm		80.02 µm		34.13 µm	
Kinetic Flame Heat Flux Density (Skeleton)	627,143.09 W/m^2		2.26e+06 W/m^2		477,668.18 W/m^2		618,751.64 W/m^2		1.71e+06 W/m^2	
Kinetic Flame Height (Skeleton)	154.7 µm		88.43 µm		133.23 µm		157.15 µm		43.96 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.25e+06 W/m^2		971,993.16 W/m^2		55,618.24 W/m^2		1.2e+06 W/m^2		3.45e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	139.23 µm		178.89 µm		786.45 µm		145.88 µm		44.29 µm	
Diffusion Flame Heat Flux Density	100,933.14 W/m^2		582,150.07 W/m^2		125,374.09 W/m^2		98,601.02 W/m^2		250,014.59 W/m^2	
Diffusion Flame Height	2,955.63 µm		523.54 µm		2,373.22 µm		3,090.61 µm		1,131.09 µm	
Metal Combustion Heat Flux Density	75,577.31 W/m^2		66,758.34 W/m^2		85,601.11 W/m^2		281,757.18 W/m^2		34,964.44 W/m^2	

2.83 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	29.12 mm/s		29.68 mm/s		20.17 mm/s		29.57 mm/s		11.03 mm/s	
Calculated Burning Rate	27.94 mm/s		28.93 mm/s		19.05 mm/s		29.2 mm/s		10.9 mm/s	
Difference	1.18 mm/s		0.75 mm/s		1.11 mm/s		0.37 mm/s		0.13 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	34.1 mm/s	13.32 mm/s	34.1 mm/s	17 mm/s	34.1 mm/s	6 mm/s	35.19 mm/s	13.95 mm/s	15.64 mm/s	5.07 mm/s
Surface Temperature	643.99 K	623.5 K	643.99 K	628.69 K	643.99 K	607.11 K	640.06 K	621.2 K	871.6 K	841.29 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	5.61e+06 W/m^2		5.61e+06 W/m^2		5.61e+06 W/m^2		5.44e+06 W/m^2		1.14e+07 W/m^2	
Kinetic Flame Height (Inter-Pocket)	54.65 μm		54.65 μm		54.65 μm		56.39 μm		24.81 μm	
Kinetic Flame Heat Flux Density (Skeleton)	987,430.98 W/m^2		3.59e+06 W/m^2		754,463.09 W/m^2		1.01e+06 W/m^2		1.71e+06 W/m^2	
Kinetic Flame Height (Skeleton)	97.98 μm		55.51 μm		84.02 μm		95.71 μm		43.72 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.5e+06 W/m^2		1.18e+06 W/m^2		66,794.63 W/m^2		1.43e+06 W/m^2		4.14e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	116.32 μm		147.76 μm		651.08 μm		122.07 μm		36.79 μm	
Diffusion Flame Heat Flux Density	88,645.96 W/m^2		517,320.44 W/m^2		110,386.42 W/m^2		86,478.73 W/m^2		220,835.18 W/m^2	
Diffusion Flame Height	3,362.26 μm		588.67 μm		2,693.16 μm		3,520.87 μm		1,279.1 μm	
Metal Combustion Heat Flux Density	90,453.42 W/m^2		78,039.45 W/m^2		102,199.8 W/m^2		338,286.84 W/m^2		41,417.79 W/m^2	

3.44 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	32.23 mm/s		33.63 mm/s		23.12 mm/s		33.9 mm/s		12.38 mm/s	
Calculated Burning Rate	31.55 mm/s		33.06 mm/s		21.73 mm/s		33.04 mm/s		12.28 mm/s	
Difference	0.68 mm/s		0.57 mm/s		1.39 mm/s		0.87 mm/s		0.098 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	42.03 mm/s	14.83 mm/s	42.03 mm/s	18.91 mm/s	42.03 mm/s	6.77 mm/s	43.37 mm/s	15.57 mm/s	19.72 mm/s	5.63 mm/s
Surface Temperature	648.73 K	625.78 K	648.73 K	630.98 K	648.73 K	609.54 K	644.48 K	623.38 K	878.12 K	844.02 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	7.4e+06 W/m^2		7.4e+06 W/m^2		7.4e+06 W/m^2		7.18e+06 W/m^2		1.47e+07 W/m^2	
Kinetic Flame Height (Inter-Pocket)	41.31 µm		41.31 µm		41.31 µm		42.65 µm		19.2 µm	
Kinetic Flame Heat Flux Density (Skeleton)	1.42e+06 W/m^2		5.16e+06 W/m^2		1.07e+06 W/m^2		1.5e+06 W/m^2		1.71e+06 W/m^2	
Kinetic Flame Height (Skeleton)	68.14 µm		38.63 µm		59.07 µm		64.5 µm		43.76 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.72e+06 W/m^2		1.35e+06 W/m^2		75,946.63 W/m^2		1.64e+06 W/m^2		4.77e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	101.08 µm		128.21 µm		569.42 µm		106.3 µm		31.87 µm	
Diffusion Flame Heat Flux Density	79,534.04 W/m^2		464,802.22 W/m^2		97,729.99 W/m^2		77,431.85 W/m^2		198,688.69 W/m^2	
Diffusion Flame Height	3,744.59 µm		654.69 µm		3,039.45 µm		3,929.42 µm		1,420.3 µm	
Metal Combustion Heat Flux Density	105,110.06 W/m^2		89,943.7 W/m^2		121,078.92 W/m^2		394,608.49 W/m^2		47,848.68 W/m^2	

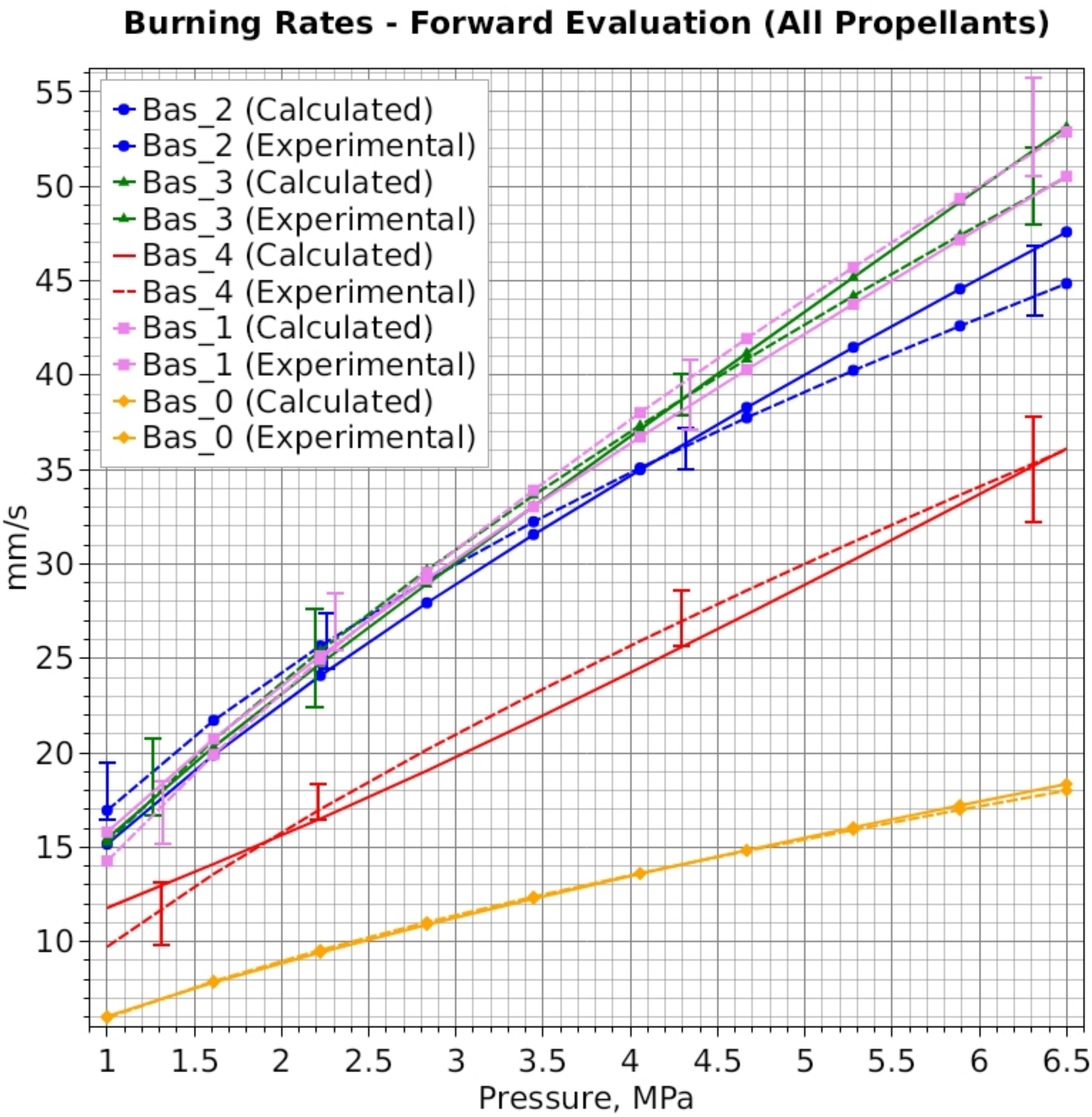
4.06 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	35.09 mm/s		37.34 mm/s		25.92 mm/s		38.01 mm/s		13.63 mm/s	
Calculated Burning Rate	34.99 mm/s		37.13 mm/s		24.49 mm/s		36.72 mm/s		13.59 mm/s	
Difference	0.1 mm/s		0.21 mm/s		1.43 mm/s		1.29 mm/s		0.036 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	50.11 mm/s	16.28 mm/s	50.11 mm/s	20.83 mm/s	50.11 mm/s	7.58 mm/s	51.72 mm/s	17.12 mm/s	23.95 mm/s	6.16 mm/s
Surface Temperature	652.77 K	627.76 K	652.77 K	633.07 K	652.77 K	611.83 K	648.25 K	625.28 K	883.64 K	846.4 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	9.33e+06 W/m²		9.33e+06 W/m²		9.33e+06 W/m²		9.05e+06 W/m²		1.82e+07 W/m²	
Kinetic Flame Height (Inter-Pocket)	32.74 μm		32.74 μm		32.74 μm		33.8 μm		15.5 μm	
Kinetic Flame Heat Flux Density (Skeleton)	1.91e+06 W/m²		6.92e+06 W/m²		1.41e+06 W/m²		2.08e+06 W/m²		1.69e+06 W/m²	
Kinetic Flame Height (Skeleton)	50.42 μm		28.75 μm		44.5 μm		46.5 μm		43.94 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	1.93e+06 W/m²		1.51e+06 W/m²		83,584.67 W/m²		1.83e+06 W/m²		5.35e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	90.11 μm		114.74 μm		514.65 μm		95.01 μm		28.36 μm	
Diffusion Flame Heat Flux Density	72,439.64 W/m²		421,685.64 W/m²		87,221.07 W/m²		70,350.19 W/m²		181,271.88 W/m²	
Diffusion Flame Height	4,108.58 μm		721.14 μm		3,403.04 μm		4,322.26 μm		1,555.45 μm	
Metal Combustion Heat Flux Density	119,628.29 W/m²		102,371.22 W/m²		141,852.22 W/m²		451,036.84 W/m²		54,239.68 W/m²	

4.67 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	37.75 mm/s		40.85 mm/s		28.6 mm/s		41.93 mm/s		14.8 mm/s	
Calculated Burning Rate	38.28 mm/s		41.17 mm/s		27.33 mm/s		40.29 mm/s		14.85 mm/s	
Difference	-0.53 mm/s		-0.32 mm/s		1.27 mm/s		1.65 mm/s		-0.044 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	58.33 mm/s	17.66 mm/s	58.33 mm/s	22.75 mm/s	58.33 mm/s	8.42 mm/s	60.22 mm/s	18.63 mm/s	28.3 mm/s	6.68 mm/s
Surface Temperature	656.3 K	629.51 K	656.3 K	635 K	656.3 K	613.96 K	651.55 K	626.98 K	888.45 K	848.51 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.14e+07 W/m^2		1.14e+07 W/m^2		1.14e+07 W/m^2		1.1e+07 W/m^2		2.19e+07 W/m^2	
Kinetic Flame Height (Inter-Pocket)	26.82 µm		26.82 µm		26.82 µm		27.7 µm		12.89 µm	
Kinetic Flame Heat Flux Density (Skeleton)	2.47e+06 W/m^2		8.87e+06 W/m^2		1.78e+06 W/m^2		2.74e+06 W/m^2		1.68e+06 W/m^2	
Kinetic Flame Height (Skeleton)	39 µm		22.42 µm		35.14 µm		35.18 µm		44.17 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.12e+06 W/m^2		1.65e+06 W/m^2		90,093.19 W/m^2		2.01e+06 W/m^2		5.9e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	81.8 µm		104.83 µm		475.1 µm		86.5 µm		25.7 µm	
Diffusion Flame Heat Flux Density	66,718.77 W/m^2		385,778.97 W/m^2		78,498.66 W/m^2		64,613.04 W/m^2		167,184.01 W/m^2	
Diffusion Flame Height	4,458.25 µm		787.76 µm		3,778.45 µm		4,703.42 µm		1,685.26 µm	
Metal Combustion Heat Flux Density	134,065.34 W/m^2		115,257.4 W/m^2		164,256.47 W/m^2		507,796.86 W/m^2		60,583.4 W/m^2	

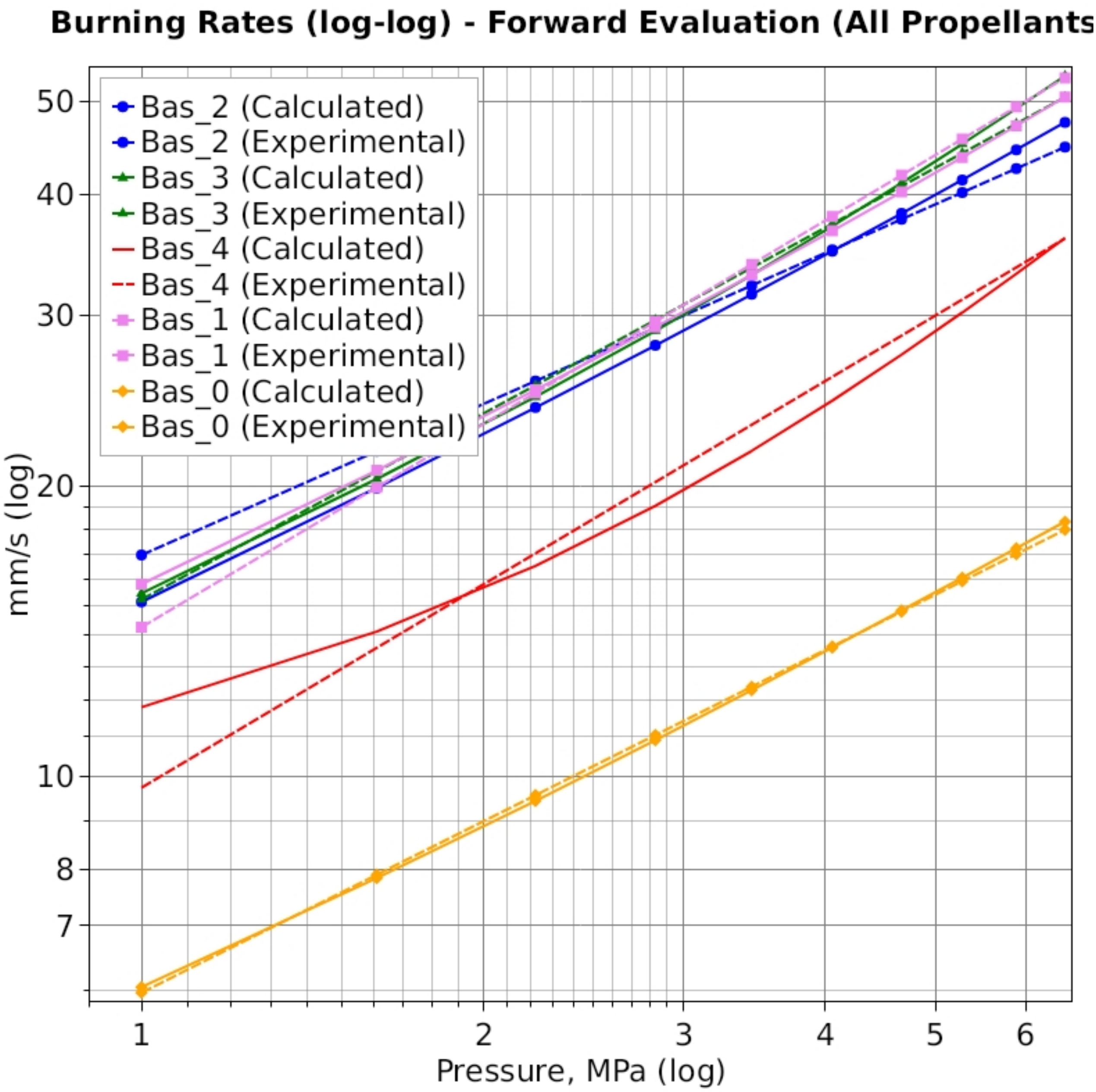
5.28 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	40.24 mm/s		44.2 mm/s		31.17 mm/s		45.71 mm/s		15.92 mm/s	
Calculated Burning Rate	41.47 mm/s		45.18 mm/s		30.22 mm/s		43.76 mm/s		16.06 mm/s	
Difference	-1.22 mm/s		-0.98 mm/s		0.95 mm/s		1.94 mm/s		-0.14 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	66.67 mm/s	19 mm/s	66.67 mm/s	24.67 mm/s	66.67 mm/s	9.27 mm/s	68.85 mm/s	20.11 mm/s	32.76 mm/s	7.17 mm/s
Surface Temperature	659.45 K	631.09 K	659.45 K	636.79 K	659.45 K	615.95 K	654.48 K	628.52 K	892.7 K	850.41 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.35e+07 W/m^2		1.35e+07 W/m^2		1.35e+07 W/m^2		1.31e+07 W/m^2		2.57e+07 W/m^2	
Kinetic Flame Height (Inter-Pocket)	22.54 µm		22.54 µm		22.54 µm		23.28 µm		10.97 µm	
Kinetic Flame Heat Flux Density (Skeleton)	3.08e+06 W/m^2		1.1e+07 W/m^2		2.18e+06 W/m^2		3.49e+06 W/m^2		1.67e+06 W/m^2	
Kinetic Flame Height (Skeleton)	31.18 µm		18.1 µm		28.7 µm		27.58 µm		44.44 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.3e+06 W/m^2		1.78e+06 W/m^2		95,745.15 W/m^2		2.18e+06 W/m^2		6.41e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	75.25 µm		97.21 µm		444.98 µm		79.82 µm		23.61 µm	
Diffusion Flame Heat Flux Density	61,982.09 W/m^2		355,467.27 W/m^2		71,211.67 W/m^2		59,843.97 W/m^2		155,528.58 W/m^2	
Diffusion Flame Height	4,796.41 µm		854.43 µm		4,162.31 µm		5,075.67 µm		1,810.34 µm	
Metal Combustion Heat Flux Density	148,462.07 W/m^2		128,558.9 W/m^2		188,107.31 W/m^2		565,046.89 W/m^2		66,877.19 W/m^2	

5.89 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	42.6 mm/s		47.41 mm/s		33.65 mm/s		49.35 mm/s		16.98 mm/s	
Calculated Burning Rate	44.56 mm/s		49.17 mm/s		33.15 mm/s		47.17 mm/s		17.22 mm/s	
Difference	-1.96 mm/s		-1.76 mm/s		0.51 mm/s		2.18 mm/s		-0.24 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	75.13 mm/s	20.3 mm/s	75.13 mm/s	26.6 mm/s	75.13 mm/s	10.14 mm/s	77.6 mm/s	21.55 mm/s	37.32 mm/s	7.65 mm/s
Surface Temperature	662.28 K	632.52 K	662.28 K	638.44 K	662.28 K	617.8 K	657.12 K	629.93 K	896.53 K	852.12 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.58e+07 W/m^2		1.58e+07 W/m^2		1.58e+07 W/m^2		1.53e+07 W/m^2		2.96e+07 W/m^2	
Kinetic Flame Height (Inter-Pocket)	19.31 µm		19.31 µm		19.31 µm		19.95 µm		9.5 µm	
Kinetic Flame Heat Flux Density (Skeleton)	3.75e+06 W/m^2		1.32e+07 W/m^2		2.59e+06 W/m^2		4.32e+06 W/m^2		1.65e+06 W/m^2	
Kinetic Flame Height (Skeleton)	25.58 µm		15 µm		24.05 µm		22.25 µm		44.71 µm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.48e+06 W/m^2		1.89e+06 W/m^2		100,733.52 W/m^2		2.33e+06 W/m^2		6.91e+06 W/m^2	
Kinetic Flame Height (Out-Skeleton)	69.95 µm		91.15 µm		421.11 µm		74.44 µm		21.91 µm	
Diffusion Flame Heat Flux Density	57,977.95 W/m^2		329,563.38 W/m^2		65,068.35 W/m^2		55,797.8 W/m^2		145,706.05 W/m^2	
Diffusion Flame Height	5,125.19 µm		921.08 µm		4,552.44 µm		5,441.2 µm		1,931.2 µm	
Metal Combustion Heat Flux Density	162,851.86 W/m^2		142,243.45 W/m^2		213,269.91 W/m^2		622,931.2 W/m^2		73,120.76 W/m^2	

6.5 MPa										
Parameter	Bas_2		Bas_3		Bas_4		Bas_1		Bas_0	
Experimental Burning Rate	44.85 mm/s		50.5 mm/s		36.06 mm/s		52.88 mm/s		18 mm/s	
Calculated Burning Rate	47.58 mm/s		53.14 mm/s		36.11 mm/s		50.53 mm/s		18.35 mm/s	
Difference	-2.73 mm/s		-2.64 mm/s		-0.043 mm/s		2.36 mm/s		-0.35 mm/s	
-	I	II	I	II	I	II	I	II	I	II
Linear Burning Rate	83.7 mm/s	21.57 mm/s	83.7 mm/s	28.52 mm/s	83.7 mm/s	11.02 mm/s	86.46 mm/s	22.98 mm/s	41.97 mm/s	8.11 mm/s
Surface Temperature	664.86 K	633.84 K	664.86 K	639.99 K	664.86 K	619.53 K	659.52 K	631.24 K	900 K	853.69 K
Kinetic Flame Heat Flux Density (Inter-Pocket)	1.81e+07 W/m²		1.81e+07 W/m²		1.81e+07 W/m²		1.76e+07 W/m²		3.36e+07 W/m²	
Kinetic Flame Height (Inter-Pocket)	16.8 μm		16.8 μm		16.8 μm		17.36 μm		8.35 μm	
Kinetic Flame Heat Flux Density (Skeleton)	4.47e+06 W/m²		1.56e+07 W/m²		3.02e+06 W/m²		5.23e+06 W/m²		1.64e+06 W/m²	
Kinetic Flame Height (Skeleton)	21.42 μm		12.69 μm		20.55 μm		18.35 μm		44.98 μm	
Kinetic Flame Heat Flux Density (Out-Skeleton)	2.64e+06 W/m²		2e+06 W/m²		105,196.13 W/m²		2.48e+06 W/m²		7.37e+06 W/m²	
Kinetic Flame Height (Out-Skeleton)	65.55 μm		86.18 μm		401.6 μm		70 μm		20.49 μm	
Diffusion Flame Heat Flux Density	54,537.29 W/m²		307,181.98 W/m²		59,838.43 W/m²		52,310.69 W/m²		137,300.66 W/m²	
Diffusion Flame Height	5,446.1 μm		987.69 μm		4,947.44 μm		5,801.42 μm		2,048.28 μm	
Metal Combustion Heat Flux Density	177,256.07 W/m²		156,287.22 W/m²		239,643.09 W/m²		681,524.9 W/m²		79,314.93 W/m²	



Burning Rates - Forward Evaluation (All Propellants)
Bas_2 (Calculated)
Bas_2 (Experimental)
Bas_3 (Calculated)
Bas_3 (Experimental)
Bas_4 (Calculated)
Bas_4 (Experimental)
Bas_1 (Calculated)
Bas_1 (Experimental)
Bas_0 (Calculated)
Bas_0 (Experimental)
mm/s
Pressure, MPa



Propellant Composition Parameters

Composition "Bas_2"

Density 1,659 kg/m³, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.

Vieille's Power Law: U = A · p^v, A = 1.285341E-05; v = 0.52.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,604 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_3"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.57.
Vieille's Power Law: U = A · p^v, A = 2.203002E-06; v = 0.64.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,674 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,623 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_4"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.56.
Vieille's Power Law: U = A · p^v, A = 6.137894E-07; v = 0.7.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,580 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,241 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,042 K

Composition "Bas_1"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.
Vieille's Power Law: U = A · p^v, A = 9.000448E-07; v = 0.7.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol

Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

Gas Phase Parameters Outside Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K

Composition "Bas_0"

Density 1,659 kg/m^3, specific heat capacity 1,500 J/kg.K, initial temperature 298 K, mass fraction of "pockets" in "binder-metal" composition 0.7.
Vieille's Power Law: U = A · p^v, A = 1.720582E-06; v = 0.59.

Gas Phase Parameters in Inter-Pocket Medium

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 3,707 K

Gas Phase Parameters in Pocket Region

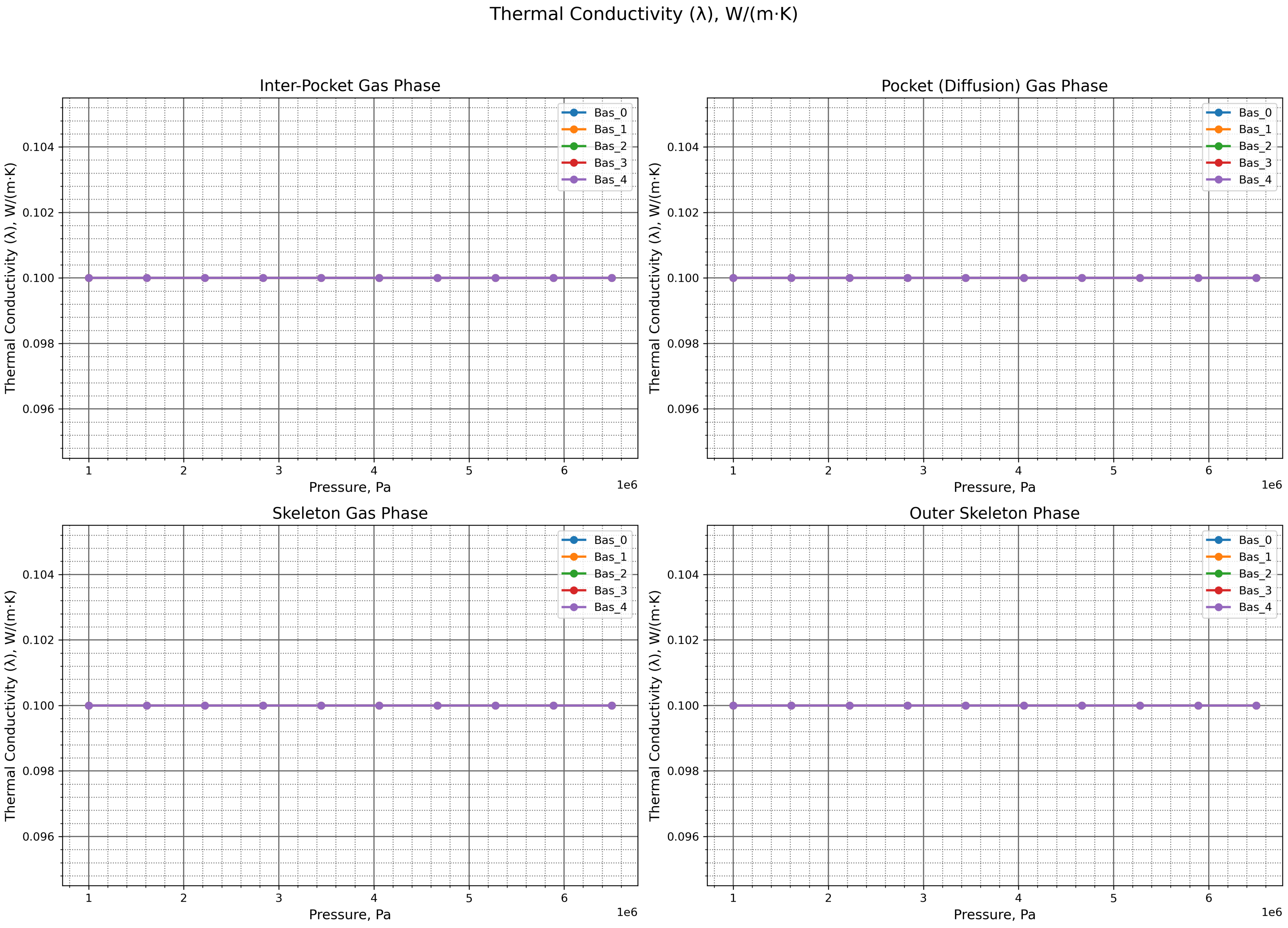
Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Diffusion flame temperature 3,666 K

Gas Phase Parameters in Skeleton Structure

Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 1,591 K

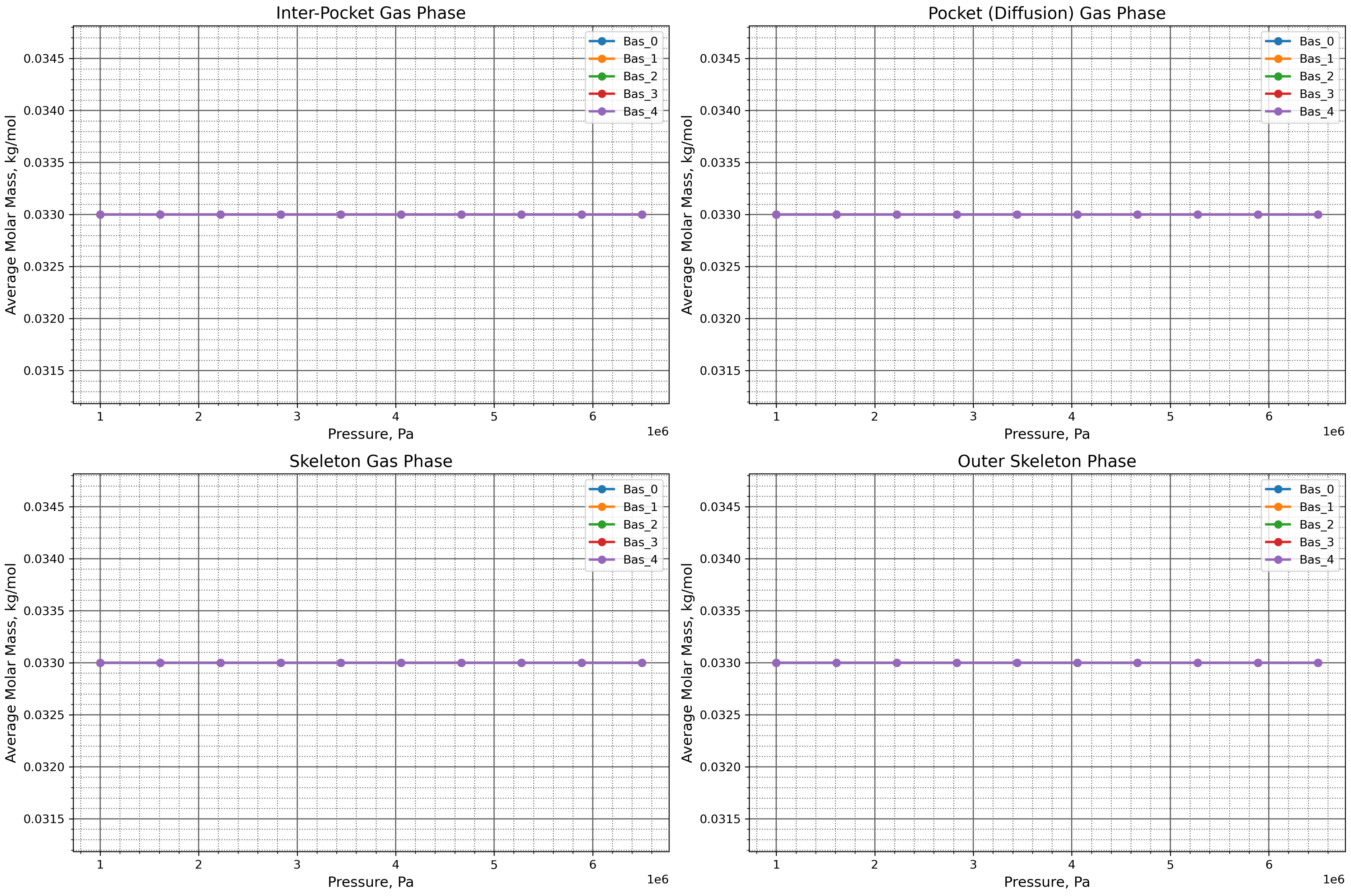
Gas Phase Parameters Outside Skeleton Structure

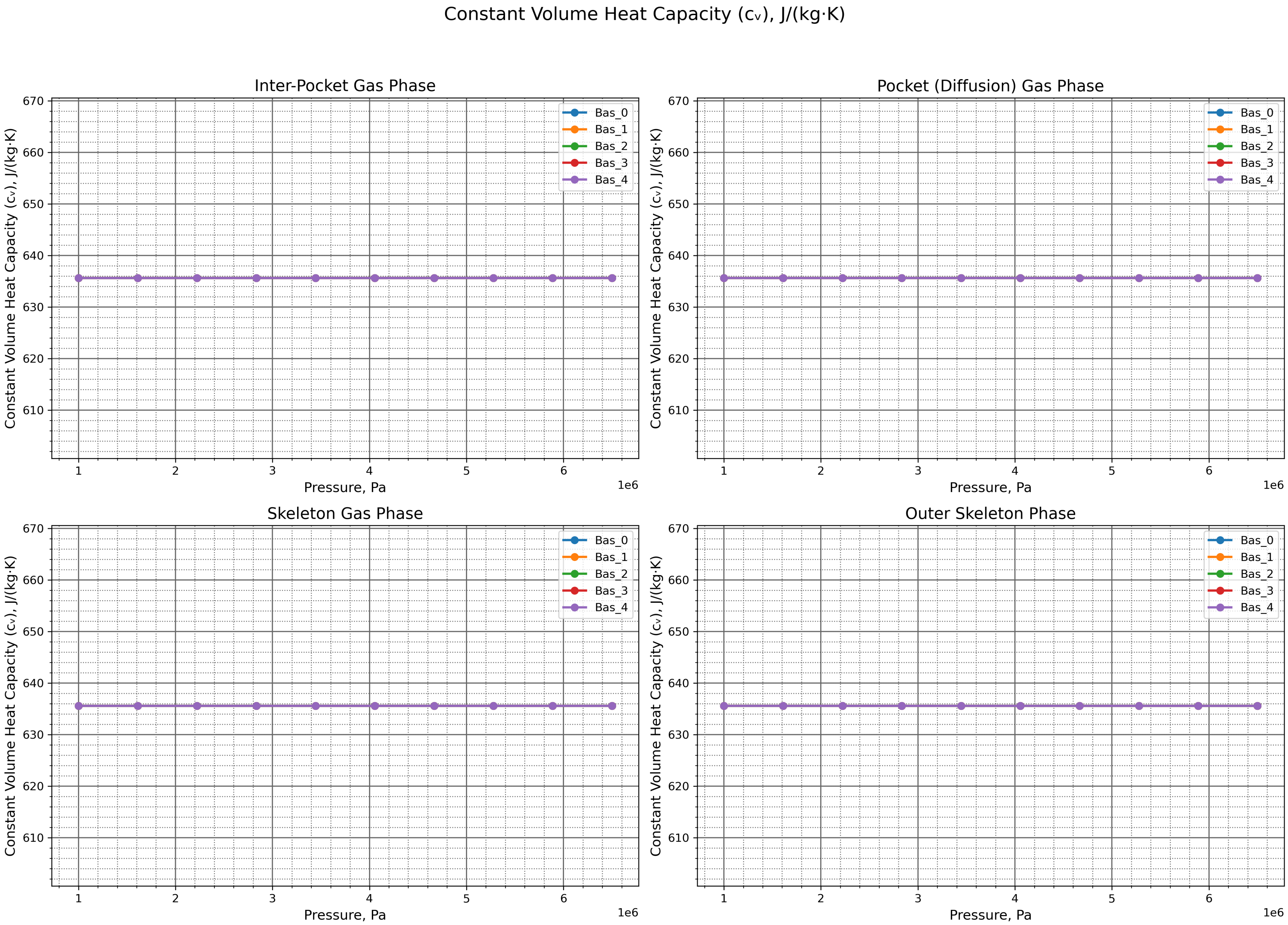
Thermal conductivity 0.1 W/(m·K)
Average molar mass 0.033 kg/mol
Specific heat capacity (constant volume) 635.6 J/kg.K
Kinetic flame temperature 2,365 K



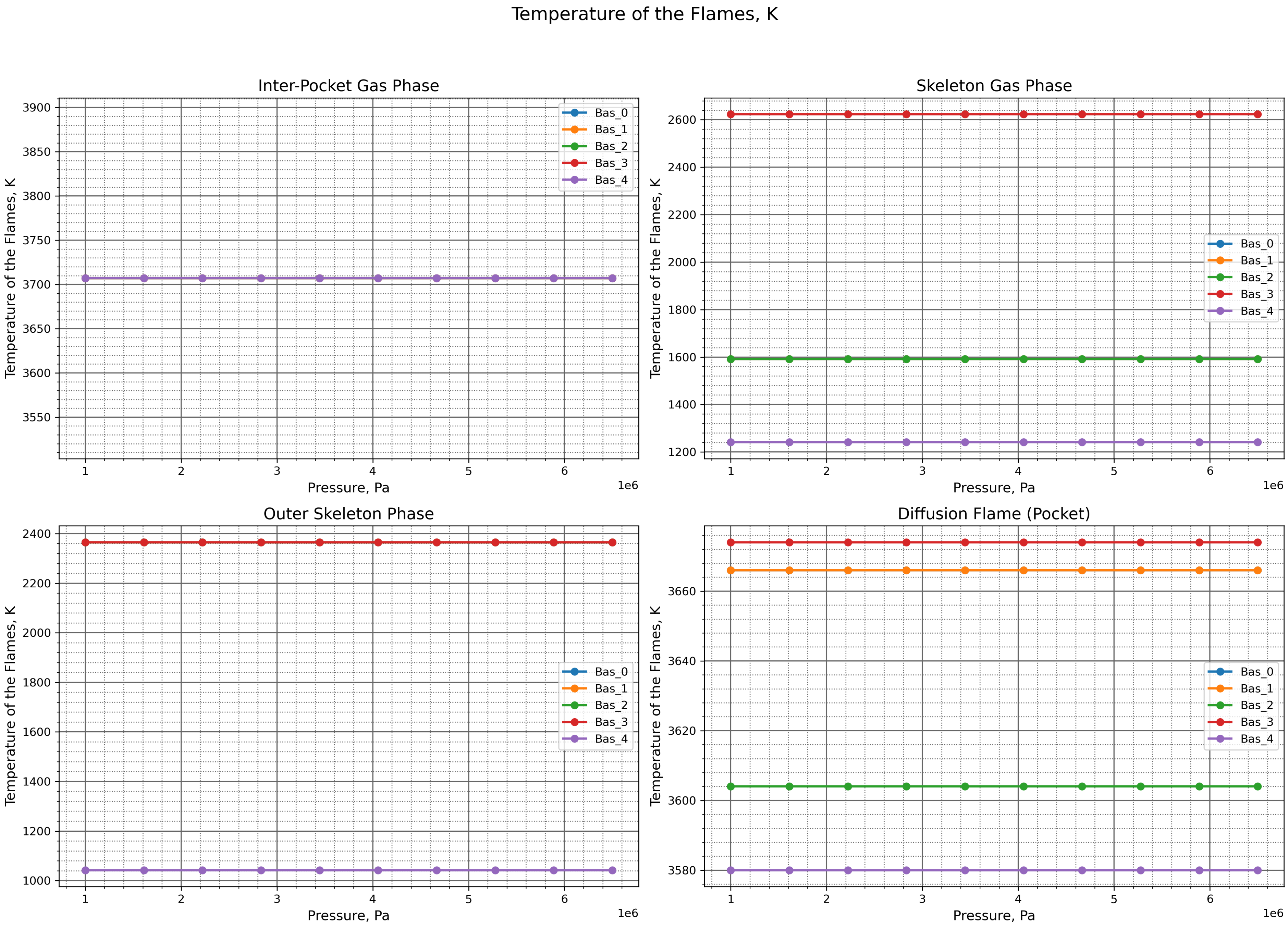
Thermal Conductivity (λ), W/(m·K)
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4
Thermal Conductivity (λ), W/(m·K)
Pressure, Pa
1e6
0.096
0.098
0.100
0.102
0.104
1
2
3
4
5
6

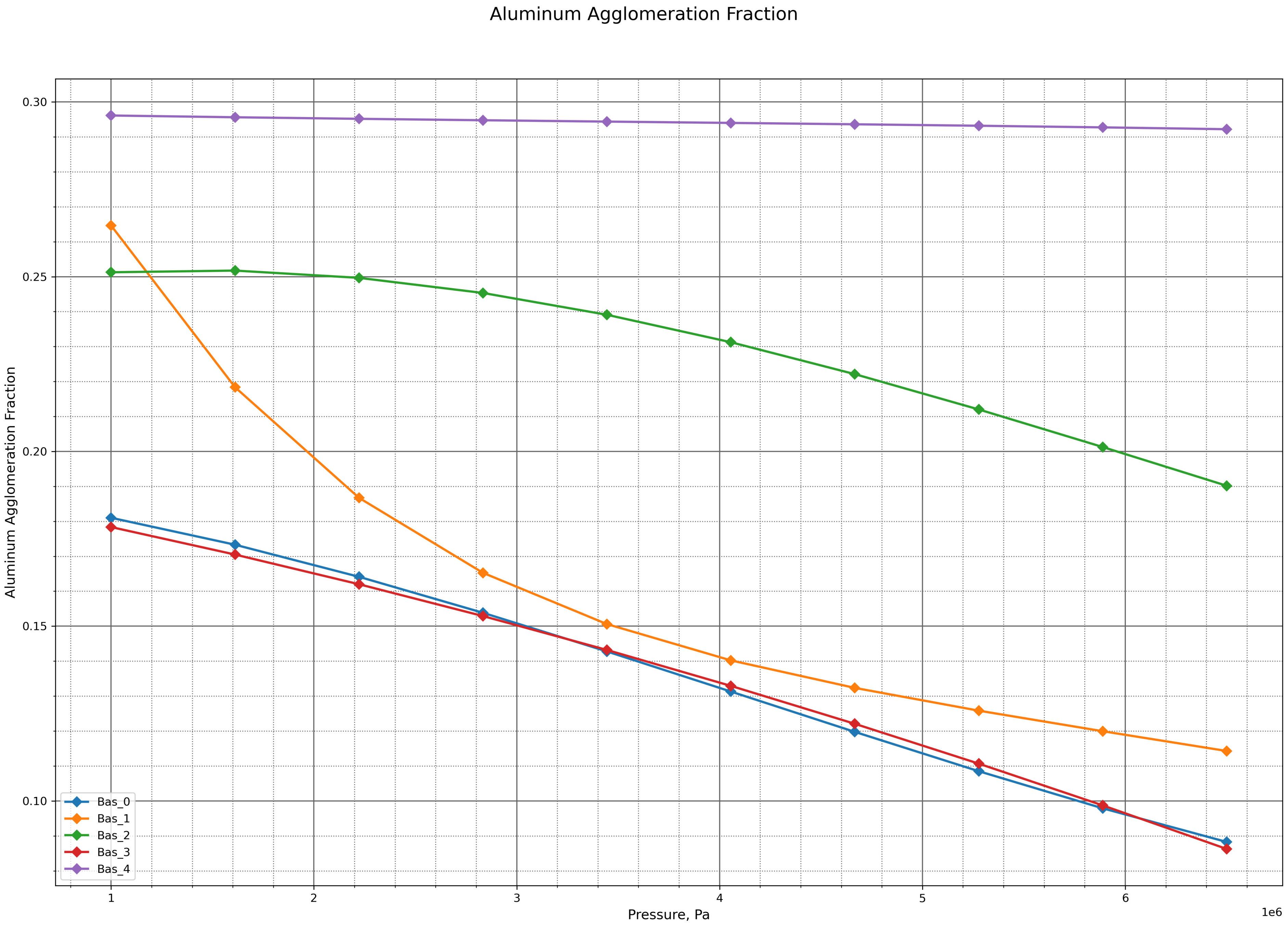
Average Molar Mass, kg/mol



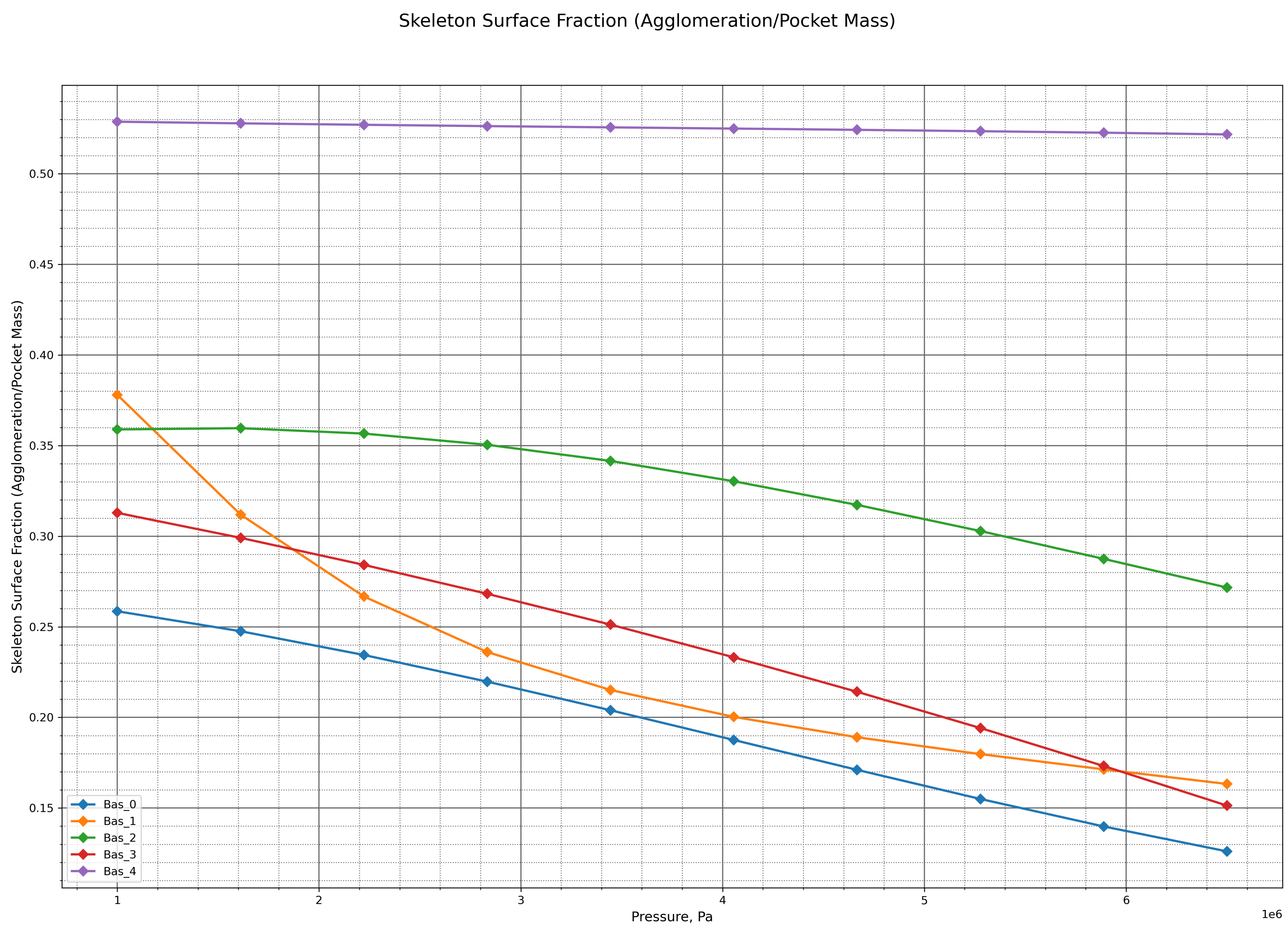


Constant Volume Heat Capacity (cᵥ), J/(kg·K)
Inter-Pocket Gas Phase
Pocket (Diffusion) Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Constant Volume Heat Capacity (cᵥ), J/(kg·K)
Pressure, Pa
1e6
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4

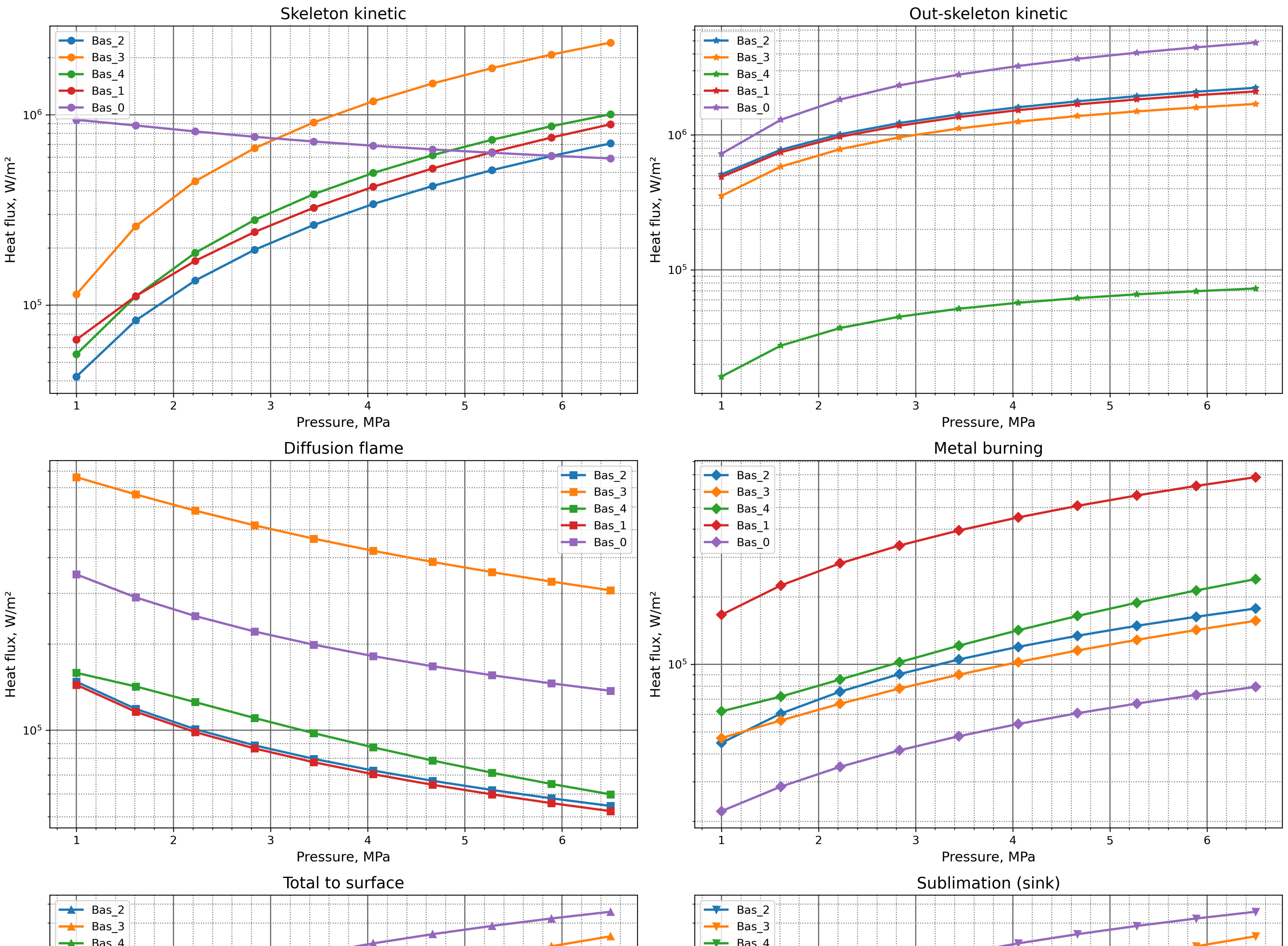

Temperature of the Flames, K
Inter-Pocket Gas Phase
Skeleton Gas Phase
Outer Skeleton Phase
Diffusion Flame (Pocket)
Temperature of the Flames, K
Pressure, Pa
1e6
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4

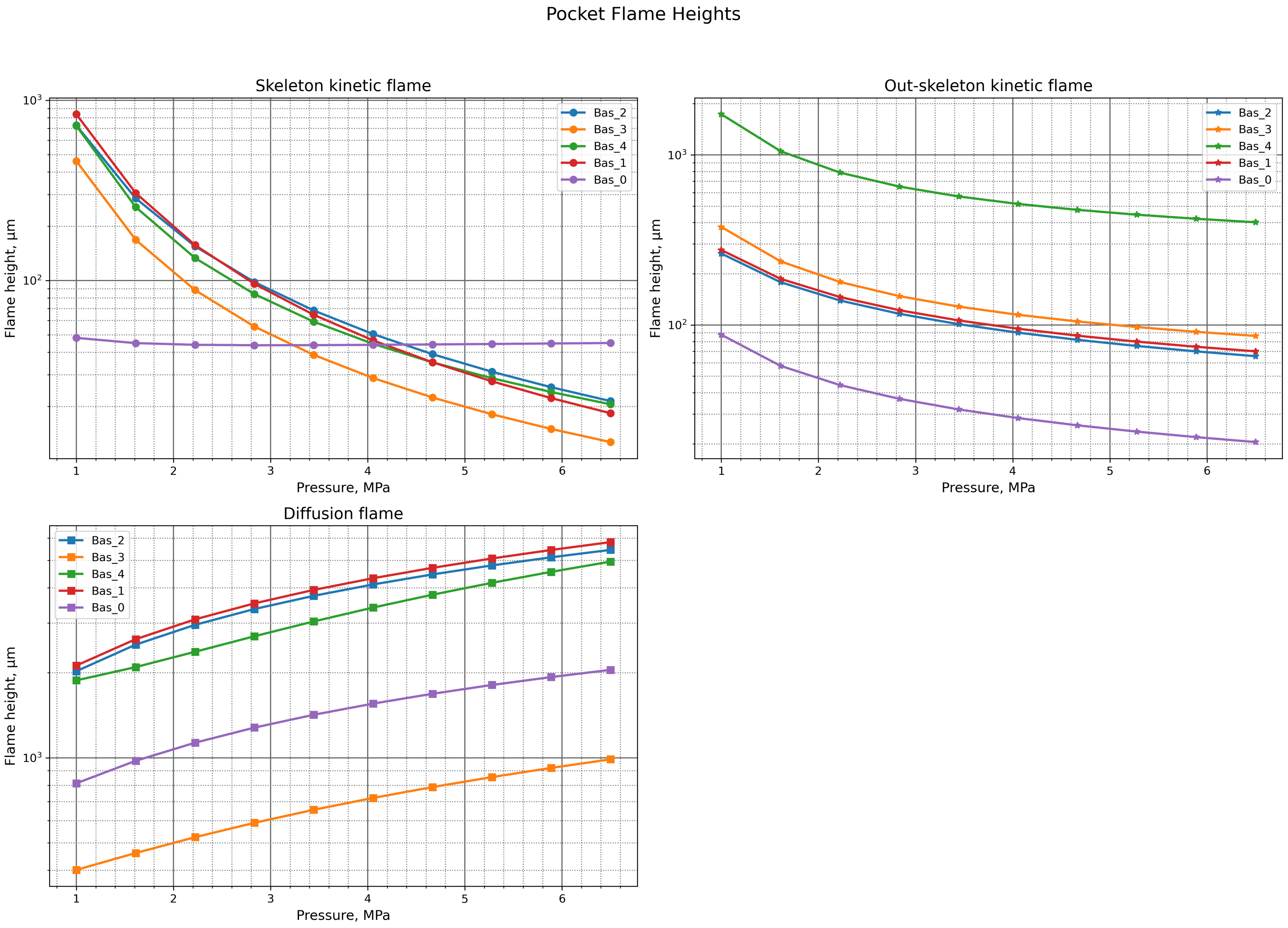


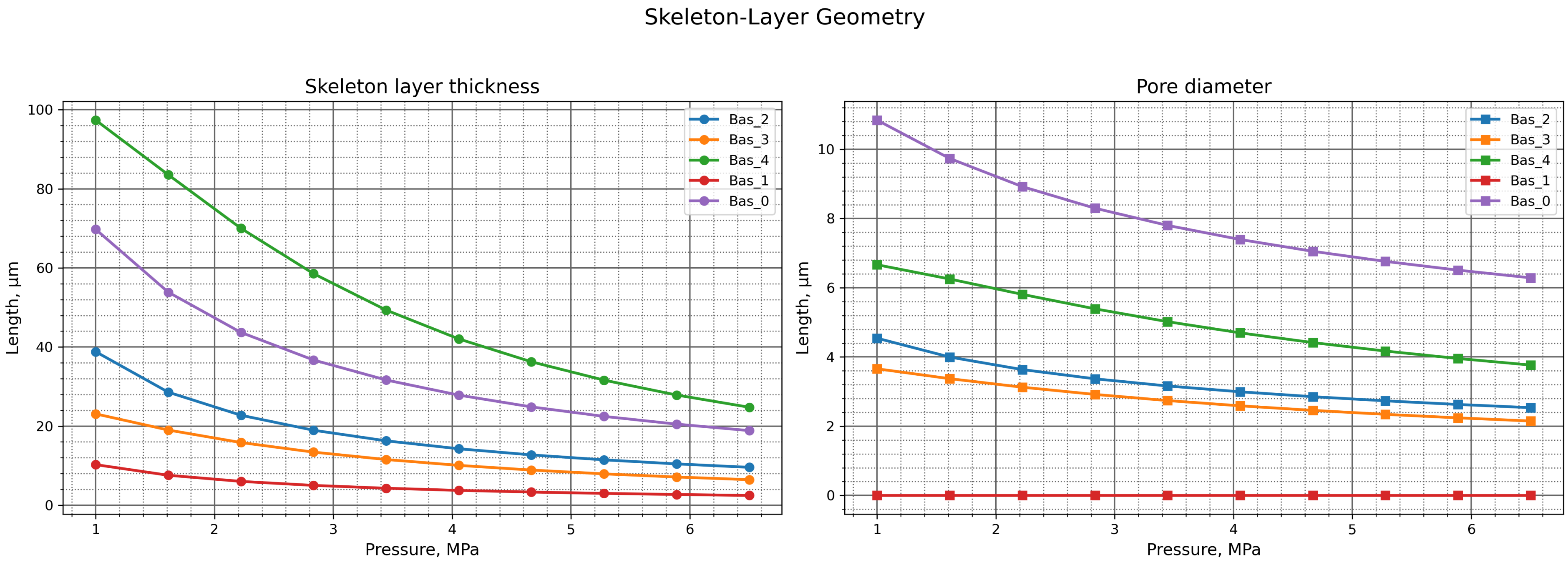
Aluminum Agglomeration Fraction
Aluminum Agglomeration Fraction
Pressure, Pa
1e6
0.30
0.25
0.20
0.15
0.10
1
2
3
4
5
6
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4


Skeleton Surface Fraction (Agglomeration/Pocket Mass)
Skeleton Surface Fraction (Agglomeration/Pocket Mass)
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
1
2
3
4
5
6
1e6
Pressure, Pa
Bas_0
Bas_1
Bas_2
Bas_3
Bas_4

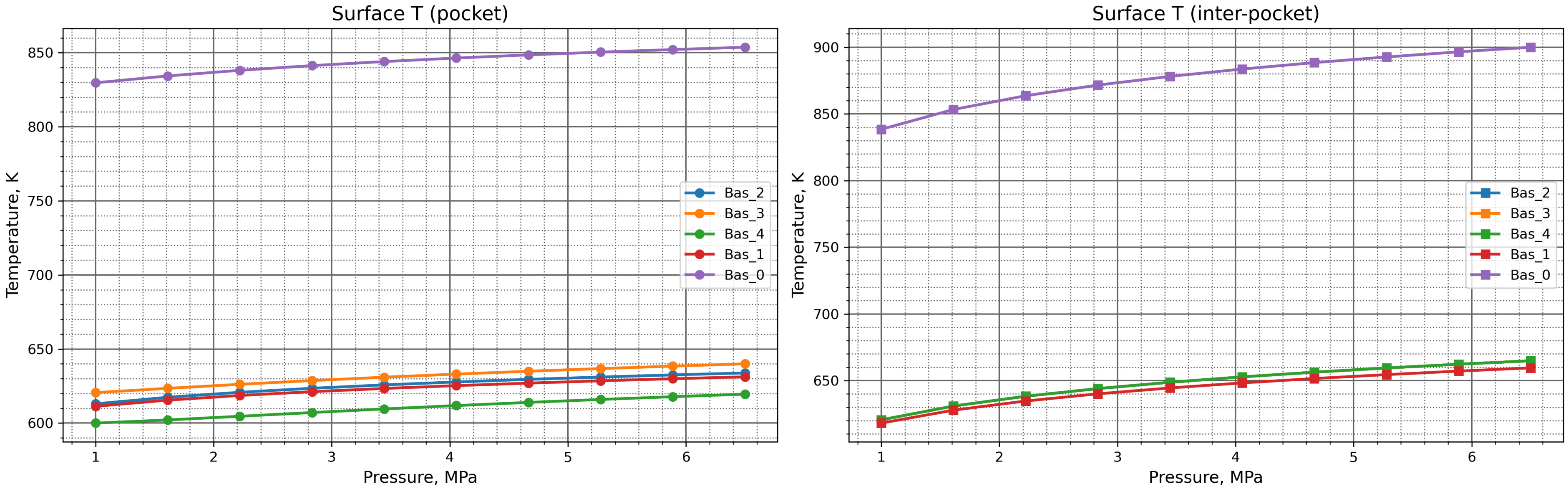
Pocket Heat-Flux Decomposition




Pocket Flame Heights
Skeleton kinetic flame
Out-skeleton kinetic flame
Diffusion flame
Bas_2
Bas_3
Bas_4
Bas_1
Bas_0
Flame height, µm
Pressure, MPa


Skeleton-Layer Geometry
Skeleton layer thickness
Length, μm
Pressure, MPa
Bas_2
Bas_3
Bas_4
Bas_1
Bas_0
Pore diameter
Length, μm
Pressure, MPa
Bas_2
Bas_3
Bas_4
Bas_1
Bas_0

Pocket Exit Temperatures



Performance Report

System Information
Operating System: Unix 6.8.0.134 (Ubuntu 24.04.4 LTS)
Processor: Intel(R) Xeon(R) CPU E5-2697 v4 @ 2.30GHz (Base Clock Frequency 2.29 GHz)
Memory: 15.6 GB (64-bit), Unknown
Machine Name: vscode
Runtime: .NET 10.0.10 (.NET 10.0.10)
Architecture: X64 / X64

Total Execution Time
Total execution time: 00:00:00.151

Garbage Collection Information
Total Memory: 12391464 bytes
GC Pause Time Percentage: 0.05%
Generation 0 Collections: 1
Generation 1 Collections: 0
Generation 2 Collections: 0

Detailed Performance Metrics
Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 0
Maximum execution time: -1.7976931348623157E+308 ms
Minimum execution time: 1.7976931348623157E+308 ms
Mean execution time: 0 ms
Execution time standard deviation: 0 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 0
Maximum execution time: -1.7976931348623157E+308 ms
Minimum execution time: 1.7976931348623157E+308 ms
Mean execution time: 0 ms
Execution time standard deviation: 0 ms

Method: Telemetry/Proc/0/OptimizationProblemExecutionPerformance
Call count: 0
Maximum execution time: -1.7976931348623157E+308 ms
Minimum execution time: 1.7976931348623157E+308 ms
Mean execution time: 0 ms
Execution time standard deviation: 0 ms